Crime Analysis Dashboard – Design & Specification Documentation

Project	CRU (Crime Response Unit) – Crime Data Visualization
Version	v1.0
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Tool	Microsoft Power BI

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1. Project Objective

This document provides a complete specification for the development of the **CRU Analytics Dashboard**, an interactive Power BI solution designed to help the **Crime Response Unit** monitor, analyze, and respond to crime cases effectively.

The goal of this project is to replicate how a real-world crime monitoring unit would function by centralizing multiple crime-related datasets into one platform, applying data cleaning and modeling techniques, and building dashboards that support **decision-making, performance tracking, and crime prevention strategies**.

It outlines the **dashboard structure, data behaviour, filters, user interactivity, applied DAX logic, and visualization design** used to create the final solution.

2. CRU (Crime Response Unit) Overview

The **Crime Response Unit (CRU)** is a specialized analytical and monitoring team concept designed to:

- **Record** crime data systematically (FIRs, victims, accused, chargesheets, disposal).
- **Analyze** crime trends by geography, type, and time period.
- **Respond** quickly to emerging issues by identifying hotspots and anomalies.
- **Support** operational teams by tracking workload, closure rates, and resolution times.
- **Improve accountability and transparency** in handling crime cases.

CRU acts as the **bridge between raw case reporting and actionable insights**, helping law enforcement agencies **plan better, deploy resources efficiently, and improve case resolution**.

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3.Target Audience

This dashboard has been designed for **police leadership, policy analysts, and senior decision-makers** in the Crime Records Unit.

- **Police Leadership (SP/DSP/PS in-charge):** To monitor performance of police stations, disposal rates, and pending backlog.
- **Policy Analysts / Crime Analysts:** To study long-term trends (YoY, seasonality, demographic breakdowns) and prepare reports for higher authorities.
- **Judicial Oversight Bodies:** To review delays in court disposal vs police disposal and identify systemic bottlenecks.
- **General Administration:** To use hotspot maps and demographic profiles for resource planning and awareness campaigns.

4. Objectives of the Dashboard

The CRU Analytics Dashboard was built with the following objectives:

- Centralize crime data into a **single, interactive platform**.
- Analyze crime patterns by **geography, time, and type of offence**.
- Support **resource planning and response strategy**.
- Improve **transparency and accountability** by tracking open vs. closed cases.
- Detect **crime hotspots, seasonal spikes, and anomalies**.
- Measure **police unit performance** (closure rate, resolution time, pending workload).

5. Data Collection

This dashboard is built on case-level data provided by the Crime Records Unit.

- **Time Range:**
  - **Earliest Record:** 31st January 2020 (01:40 AM)
  - **Latest Record:** 30th January 2025 (11:50 PM)
  - **Total Span:** ~5 years of data (Feb 2020 – Jan 2025)
  - **Complete Years:** 2020, 2021, 2022, 2023, 2024
  - **Partial Year:** 2025 (only data until January 30, 2025). Treated as **YTD** and excluded from YoY comparisons to avoid misleading trends.
- **Geographical Coverage:** Data spans across **45 districts and 50 police stations**.
- **Data Grain:** Case-level (each row represents one FIR/case linked with CaseMaster and associated tables).
- **Data Volume:**

Table Name	Records	Columns	Description
Accused	237,773	15	Contains demographic and identifying details of accused individuals, including age, gender, caste, and occupation, which enable profiling and trend analysis.
Chargesheet	54,325	6	Provides information about chargesheets filed, such as chargesheet number, filing date, and linking with case ID, helping to measure timeliness and compliance.

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Act Section	349,029	6	Captures the sections of law under which cases are registered. This helps in classifying crimes into Heinous/Non-Heinous and analyzing crime categories under IPC and Special Acts.
Crime	211,863	18	Core dataset containing FIR and case-level details such as crime type, location, registration date, and linkage with police station IDs. Serves as the backbone of the analysis.
Disposal	17,660	5	Stores case disposal information, including disposal type (e.g., Convicted, Acquitted, Pending) and disposal date, which supports efficiency and backlog analysis.
PS Unit	1,856	4	Master reference table listing all police stations, districts, and geographical coordinates (latitude/longitude) used for spatial mapping and hotspot analysis.
Victim	141,398	15	Contains demographic details of victims, including age, gender, caste, and occupation. Used to identify vulnerable groups and generate victim demographic profiles.

6. Data Explanation (Data Dictionary)

6.1 Crime Table (Main Case Information)

Record: FIR (First Information Report) details

Field	Description	Use Case
District_Name	District where the crime was reported	Important to see which districts have higher/lower crime.
UnitName	Police station/unit handling the case.	Helps analyze workload across stations.
FIRNo	FIR number (official police case number)	Unique identifier used internally.
Fir_ID	System-generated ID for each FIR.	Used to link with other tables (Victim, Accused, etc.).
Pin_Code	Area postal code	Gives location context for mapping.
Direction_Distance	Location description relative to a landmark	Helps in tracing exact spot if GPS is missing

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Beat	Police beat (smaller area under a station)	Used for micro-level hotspot analysis.
Village	Village name	
FIR Type	Cognizable / Non-Cognizable offence	Cognizable = police can arrest without warrant (e.g., murder, theft). Non-cognizable = need court approval
FIR_Stage	Stage of FIR (e.g., Registered, Under Investigation, Closed)	Tracks case progress
CrimeGroup_Name	Broad category of crime (Theft, Assault, Cybercrime)	
CrimeHead_Name	Specific type (e.g., Vehicle Theft, Chain Snatching)	
Latitude & Longitude	Geo coordinates of crime spot.	For mapping & hotspot detection.
Offence_From_Date / To_Date	Period when crime occurred.	
FIR_Date	Date FIR was registered	Date FIR was registered.

6.2 Victim Table

Record: Stores information about victims

Field	Description	Use Case
Sex	Gender of victim (Male/Female/Other)	Important for gender-based crime studies
Age / DOB	Age & Date of Birth of victim	Used for age-based vulnerability analysis (e.g., crimes against children, elderly)
Height	FIR number (official police case number)	
Nationality, Religion, Caste	Background detail	Sociological analysis, but must be anonymized for privacy.
Occupation	Job of victim.	Detects if certain groups (students, workers,

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		professionals) are targeted more
Perm_Pincode / Pres_Pincode	Permanent & present addresses	
Injury_Type	Type of injury (e.g., minor, grievous)	
Injury_Nature	Nature of harm (e.g., physical, psychological)	
VictimType	Individual, Institution, Company etc	
FIR_ID, Unit_ID	Links victim to the case	

6.3 Accused Table

Record: Information about the accused persons.

Field	Description	Use Case
Arr_ID	Unique ID of accused person in dataset	
OffenderType	First-time offender, habitual offender, etc	
Sex, Age, DOB	Demographics of accused	
Height, Nationality, Occupation, Religion, Caste	Background info	
Conviction_Date	If convicted, date of conviction	Useful for conviction rate tracking
Perm_Pincode, Pres_Pincode	Addresses of accused	
FIR_ID, Unit_ID	Connects accused to FIR	

6.4 ChargeSheet Table

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Record: When investigation is complete, police file a **chargesheet** in court.

Field	Description	Use Case
CaseMasterID	Case identifier	for linking with Dispose table
CASEChargesheetID	Chargesheet unique ID	
ChargesheetNo	Official chargesheet number	
Fir_id	Connects to main crime case	
Unit_id	Police unit handling the case	
Chargesheet_Date	Date when chargesheet was filed	

6.5 Dispose Table

Record: Represents how and when cases were disposed/closed.

Field	Description	Use Case
CaseMasterID	Unique case identifier	Links disposal records with case master table
CASEChargesheetID	Chargesheet unique ID	Connects disposal record with chargesheet
Fir_id	FIR identifier	Links disposal entry to main case FIR
Unit_id	Police unit ID	Identifies which police unit handled the case
Dispose_Date	Date of case disposal/closure	Used for calculating case resolution and pending duration

6.6 Act Section Table

Record: Details of legal acts/sections applied to cases.

Field	Description	Use Case

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Act_Code	Legal Act code (e.g., IPC, IT Act)	Identifies the main Act under which case was filed
Act_Description	Full name of Act	Provides clarity on Act used (e.g., Indian Penal Code)
Section_Code	Section number of Act (e.g., IPC 302)	Specifies exact section invoked (e.g., Murder)
Sec_Description	Section details/definition	Explains scope of legal section
Fir_id	FIR identifier	Connects applied Act/Section to a specific FIR
Unit_id	Police unit ID	Identifies police unit responsible for applying section

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6.7Ps Unit Table (Police Station Master)

Record: Lookup table for police stations.

Field	Description	Use Case
UnitID	Unique police station identifier	Primary key to link cases with specific police station
UnitName	Name of the police station	For reporting and station-level analysis
Latitude	Latitude coordinate of station	Enables mapping police stations on GIS/Spatial visuals
Longitude	Longitude coordinate of station	Used along with latitude to plot PS locations on map
UnitID	Unique police station identifier	Primary key to link cases with specific police station
UnitName	Name of the police station	For reporting and station-level analysis

7. Power Query Transformation

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7.1 Fact_Crime

Step	Description	Action	Reason (Why + How it helps)
1	Imported dataset from Excel	Import Data	Ensures full raw Fact_Crime dataset is available for transformation
2	Selected sheet "Crime"	Select Sheet	Focuses only on crime records → avoids loading irrelevant sheets
3	Promoted first row as headers	Promote Headers	Converts row into usable field names → improves readability
4	Converted fields into correct data types	Change Data Type	Prevents calculation errors, supports consistent reporting
5	Created Key column = Fir_id & Unit_ID	Add Custom Column	Ensures unique identifiers → supports accurate joins
6	Extracted Year from FIR_Date	Extract Function	Enables yearly trends & YOY analysis
7	Renamed FIR_Date → FIR_Date_Time	Rename Column	Differentiates datetime field from date-only version
8	Duplicated FIR_Date_Time	Duplicate Column	Backup for transformation into clean date-only
9	Renamed FIR_Date_Time - Copy → FIR_Date	Rename Column	Restores dedicated date field for analysis
10	Converted FIR_Date to date-only	Change Data Type	Ensures clean joins with Calendar & trends
11	Created Case Status column (Active/Closed/Blank)	Conditional Column	Simplifies FIR stages → helps in workload monitoring
12	Extracted time from Offence_From_Date	Extract Function	Supports time-of-day crime analysis
13	Converted Offence Time → Time type	Change Data Type	Validates time field → enables grouping
14	Created Offence Time Period buckets	Conditional Column	Groups crimes by dayparts → helps spot peak hours
15	Converted text fields to Proper Case	Text Transformation	Removes inconsistencies → avoids duplicates
16	Duplicated Longitude column	Duplicate Column	Keeps original safe → one copy for mapping

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17	Renamed Latitude/Longitude copies → Lat, Long	Rename Columns	Short names → easier for map visuals
18	Converted FIR_Date → DateTime	Change Data Type	Supports calculations needing timestamps
19	Merged Fact_Crime with ChargeSheet table	Merge Query	Adds chargesheet details → links lifecycle data
20	Extracted Chargesheet_Date	Expand Column	Keeps only necessary column for analysis
21	Renamed Chargesheet_Date	Rename Column	Standard, intuitive naming
22	Merged Fact_Crime with Dispose table	Merge Query	Adds disposal information
23	Extracted Dispose_Date	Expand Column	Keeps only relevant column
24	Renamed Dispose_Date	Rename Column	Consistency across model
25	Added column = Chargesheet_Date - FIR_Date	Add Custom Column	Tracks investigation time (FIR → Chargesheet)
26	Converted investigation time → Whole Number	Change Data Type	Numeric consistency → enables averages
27	Added column = Dispose_Date - FIR_Date	Add Custom Column	Tracks total case resolution timeline
28	Converted disposal time (DD - CD) → Whole Number	Change Data Type	Ensures numeric reliability
29	Added column = Dispose_Date - FIR_Date	Add Custom Column	Alternative KPI for disposal duration
30	Converted disposal time (DD - FD) → Whole Number	Change Data Type	Provides clean closure KPI
31	Created Case_status column with 4 groups	Conditional Column	Consolidates multiple FIR stages → supports funnel/lifecycle views
32	Extracted Hour from Offence Time	Extract Function	Enables hour-level crime analysis
33	Converted FIRNo → Text	Change Data Type	Preserves FIR formatting (leading zeros)
34	Removed duplicates on Key (2 rounds)	Remove Duplicates	Ensures one record per FIR → avoids double counting

7.2 Dim_Accused

Step	Description	Action	Reason (Why + How it helps)

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1	Source Import	Connected Power Query to Excel file final_dataset_for_showcase.xlsx and selected **Accused** sheet	Brings raw accused records into Power BI → starting dataset with demographic & legal details
2	Select Sheet	Selected **Accused** sheet from workbook	Loads only relevant sheet → avoids unnecessary tables
3	Promote Headers	Converted first row to column headers	Provides proper field names (Arr_ID, OffenderType, etc.) → prevents generic names like Column1
4	Change Data Types	Converted: Arr_ID, Height, Age → Whole Number; DOB, Conviction_Date → Text (initial); OffenderType, Sex, Religion, Occupation, Caste, Nationality, Unit_ID, FIR_ID → Text	Ensures numeric/date/category fields behave correctly → avoids calculation errors in averages, timelines, groupings
5	Replace NULL in Demographics	Replaced "NULL" with "Unknown" in Sex, Religion, Caste	Standardizes missing values → prevents NULL showing in visuals, keeps category grouping clean
6	Create Case Key	Added **Key = FIR_ID & "-" & Unit_ID**	Unique case-level key → ensures consistent joins across tables
7	Create Age Segment	Conditional Column: If Age=0 → Unknown; ≤12 → Children (0–12); ≤19 → Teenagers (13–19); ≤29 → Youths (20–29); ≤39 → Adults (30–39); ≤59 → Middle-Aged (40–59); Else → Senior Citizen (60+)	Groups raw ages into categories → simplifies demographic analysis
8	Standardize Text to UPPER	Converted Sex, Religion, Caste → UPPERCASE	Removes variations (male/Male/MALE) → prevents duplicate categories
9	Format to Proper Case	Converted Sex, Religion, Caste, OffenderType, Occupation → Proper Case	Improves readability → professional-looking labels in visuals
10	Create Offender Key	**Added** Key_Offender = FIR_ID & "-" & Unit_ID & "-" & Arr_ID	Guarantees unique identifier per accused → enables headcount analysis
11	Standardize Sex Codes	Conditional Column: M/m → Male; F/f → Female; T/t → Trans; N/n → Unknown; else → Unknown	Cleans inconsistent codes → supports reliable gender-based crime analysis

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12	Filter Invalid Ages	Filtered rows where Age >= 0 and Age <= 100	Removes unrealistic ages → ensures trustworthiness in averages & demographics
13	Replace NULL in Occupation	Replaced “Null” with “Unknown” in Occupation	Consistent handling of missing values → prevents fragmented categories
14	Convert Conviction Date	Changed Conviction_Date from Text → DateTime	Enables timeline calculations → supports KPIs like time-to-conviction

7.3 Dim_act_section

Step	Description	Action	Reason (Why + How it helps)
1	Source Import	Imported **Crime** sheet from Excel file	Provides base case-level dataset
2	Promote Headers	Converted first row to column headers	Standardizes structure → prevents Column1, Column2 issues
3	Change Data Types	FIR ID, Unit ID → Text; Dates → DateTime; numeric fields → Whole Number	Ensures accuracy in joins & calculations
4	Create Case Key	Added **Key = FIR_ID & "-" & Unit_ID**	Unique identifier per case → enables joins with Dispose & Chargesheet

7.4 Dim_ chargesheet

Step	Description	Action	Reason (Why + How it helps)
1	Source Import	Imported **Chargesheet** sheet from Excel file	Provides chargesheet-level data for FIR tracking
2	Select Sheet	Selected **Crime** sheet from workbook	Avoids unnecessary sheets, loads only relevant data
3	Promote Headers	Converted first row to headers	Standardizes structure → clean field names
4	Change Data Types	IDs → Int64; FIR & Unit → Text; Date → DateTime	Ensures correct formats → avoids type errors in joins
5	Create Case Key	Added **Key = Fir_id & "-" & Unit_id**	Uniquely identifies FIR across units → reliable joins

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6	Group & Aggregate	Grouped by Key → took **List.Max(Chargesheet_Date)**	A case can have multiple chargesheets → ensures analysis uses latest date

7.5 Dim_Disposed

Step	Description	Action	Reason (Why + How it helps)
1	Source Import	Imported Dispose sheet from Excel file	Provides case disposal information → enables tracking of FIR outcomes
2	Select Sheet	Selected Crime sheet from workbook	Avoids unnecessary sheets, loads only relevant data
3	Promote Headers	Converted first row to headers	Standardizes structure → provides clean column names
4	Create Case Key	Added Key = Fir_id & "-" & Unit_id	Ensures unique FIR across units → supports joining with fact_crime
5	Change Data Type (Dispose Date)	Converted Dispose_Date → DateTime (en-IN locale)	Standardizes dates → enables time-based analysis
6	Change Data Types (IDs)	Converted IDs (CaseMasterID, CASEChargesheetID) → Int64; FIR & Unit → Text	Ensures consistency in joins & calculations
7	Remove Duplicates	Kept distinct rows on CaseMasterID	Removes repeated entries → prevents double-counting disposed cases

7.6 dim_ps_lat_long

Step	Description	Action	Reason (Why + How it helps)
1	Source Import	Imported Ps Unit sheet from Excel file	Extracts police station details with latitude & longitude → adds geospatial dimension
2	Select Sheet	Selected Crime sheet from workbook	Loads only relevant data → avoids unnecessary sheets/tables
3	Promote Headers	Converted first row to headers	Standardizes structure → provides clean field names for joins

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4	Change Column Types	Converted UnitID → Int64; UnitName → Text; Latitude & Longitude → Decimal Numbers	Ensures correct formats → enables mapping and numeric operations
5	Standardize Text	Converted all UnitName values to UPPERCASE	Removes variations in casing → prevents mismatches during joins
6	Remove Duplicates	Removed duplicates based on UnitName	Maintains one unique police station per coordinate → ensures accurate mapping

7.7 Dim_Victim

Step	Description	Action	Reason (Why + How it helps)
1	Source Import	Imported Victim sheet from *final_dataset_for_showcase.xlsx*	Extracts victim-related details → builds victim dimension linked to FIRs
2	Select Sheet	Selected Victim sheet from workbook	Ensures only relevant raw data is loaded → avoids unnecessary sheets/tables
3	Promote Headers	Promoted first row to headers	Standardizes structure → provides proper column names for transformations
4	Change Column Types	Converted fields: Sex → Text; Age, Height → Int64; DOB → Date; FIR_ID & Unit_ID → Text; others to Text	Correct formats ensure: numeric (Age, Height) usable in calcs; DOB for time analysis; text fields for grouping
5	Remove Unnecessary Columns	Dropped DOB & Perm_Pincode	Reduces noise → keeps table focused on essential fields
6	Replace Nulls	Replaced NULL with “Unknown” in Sex, Occupation, Religion, Caste, Injury_type	Standardizes missing data → avoids broken aggregations and ensures categorization
7	Remove Columns (Injury)	Removed Injury_type & Injury_Nature	Simplifies dimension → keeps focus on demographics and victim type
8	Create Key Column	Added Key = FIR_ID & "-" & Unit_ID	Ensures uniqueness across units → supports joins with fact_crime and timeline analysis
9	Create Age Segmentation	Added Age Segment categories (Children, Teenagers, Youths, Adults,	Groups victims into brackets → enables

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		Middle-Aged, Senior Citizen, Unknown)	demographic profiling & comparative analysis
10	Standardize Text	Applied Text.Upper / Text.Proper across Caste, Occupation, Religion, Sex, VictimType	Handles inconsistent casing → prevents duplicates and ensures clean categories
11	Rename Column	Renamed Sex → Sex_v1	Keeps original field before recoding → avoids overwriting raw values
12	Recode Sex	Created standardized Sex column (Male/Female/Trans/Unknown)	Cleans inconsistent entries → enables accurate gender-based victim analysis
13	Filter Age Range	Kept rows where Age ∈ [0,100]	Removes outliers → ensures demographic analysis is realistic

7.8 Victim Age Segment Order Table

Step	Description	Action	Reason (Why + How it helps)
1	Source Import (Manual Table)	Created custom table with victim age groups: Children (0–12), Teenagers (13–19), Youths (20–29), Adults (30–39), Middle-aged (40–59), Senior Citizen (60+), Unknown	Defines clear demographic brackets → ensures victims are grouped logically
2	Add Index Column	Added sequential index values (1–7)	Provides ordering mechanism for categories
3	Rename Index Column	Renamed Index → Order	Order column ensures custom sort sequence is preserved
4	Custom Sorting	Used Order column for sorting age groups	Displays age progression naturally (Child → Teenager → Youth → Adult → Middle-aged → Senior) instead of alphabetical order → supports clearer demographic trend analysis

7.9 Victim Sex Order Table

Step	Description	Action	Reason (Why + How it helps)
1	Source Import (Manual Table)	Created custom table with Victim Gender categories: Male,	Defines clear categories for gender dimension

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		Female, Trans, Unknown	
2	Add Index Column	Added sequential index values (1–4)	Provides ordering mechanism for categories
3	Rename Index Column	Renamed **Index → Order**	Order column enables consistent sorting
4	Custom Sorting	Applied **Order** column for gender categories	Ensures visuals display genders in logical sequence (Male → Female → Trans → Unknown), rather than alphabetical → improves readability in charts, slicers, and demographic trends

7.10 Offender Sex Order Table

Step	Description	Action	Reason
1	Source Import (Manual Table)	Created custom table with Offender Gender categories: Male, Female, Trans, Unknown	Defines consistent offender gender categories
2	Add Index Column	Assigned sequential index values (1–4)	Creates logical ordering mechanism
3	Rename Index Column	Renamed **Index → Order**	Prepares column for use in custom sort
4	Custom Sorting	Applied **Order** column for offender gender categories	Maintains sort consistency with victim gender → allows side-by-side comparison of victim vs offender gender distribution in visuals

7.11 Offender Age Segment Order Table

Step	Description	Action	Reason
1	Source Import (Manual Table)	Created custom table with Offender Age Segments: Children (0–12), Teenagers (13–19), Youths (20–29), Adults (30–39), Middle-aged (40–59), Senior Citizen (60+), Unknown	Establishes clear offender age categories
2	Add Index Column	Assigned sequential numbers (1–7)	Creates a natural chronological order
3	Rename Index Column	Renamed Index → Order	Prepares field for custom sorting
4	Custom Sorting	Applied Order column for offender age categories	Ensures offender age groups appear in chronological order, making crime pattern analysis by age group accurate and meaningful

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7.12 Time Order Table

Step	Description	Action	Reason
1	Source Import (Manual Table)	Created custom table with Time Segments: Midnight–6 AM, 6 AM–12 PM, 12 PM–6 PM, 6 PM–Midnight	Defines logical daily time buckets for crime trend analysis
2	Add Index Column	Assigned sequential numbers (1–4)	Establishes natural progression of time
3	Rename Index Column	Renamed Index → Order	Used for custom sort order
4	Custom Sorting	Applied Order column to time segments	Ensures time slots are arranged chronologically instead of alphabetically
5	Analysis Impact	Enables crime distribution analysis by time of day (e.g., morning vs night)	Helps identify high-risk time periods for preventive actions

7.13 Calendar Table

Code Used:

```
Calender =
VAR startdate = DATE(YEAR(MIN(Fact_Crime[FIR_Date_Time])),01,01)
VAR Enddate = DATE(YEAR(MAX(Fact_Crime[FIR_Date_Time])),12,31)
RETURN
   ADDCOLUMNS(
              CALENDAR(startdate,Enddate),
              "year",YEAR([Date]),
              "month",MONTH([Date]),
              "month name",FORMAT([Date],"mmm"),
              "day",FORMAT([Date],"dddd"),
              "day num",WEEKDAY([Date],1),
              "quater",QUARTER([Date]),
              "year month",FORMAT([Date],"YYYY-MM"),
              "week",WEEKNUM([Date],2),
              "halfyearstart",FORMAT([Date],"MM yyyy"),
              "halfyearsort",YEAR([Date])*100+MONTH([Date]))
```

Step	Description	Action	Reason
1	Define Date Range	Used MIN(Fact_Crime[FIR_Date_Time]) as start date and MAX(Fact_Crime[FIR_Date_Time]) as end date	Ensures calendar covers all available crime data period

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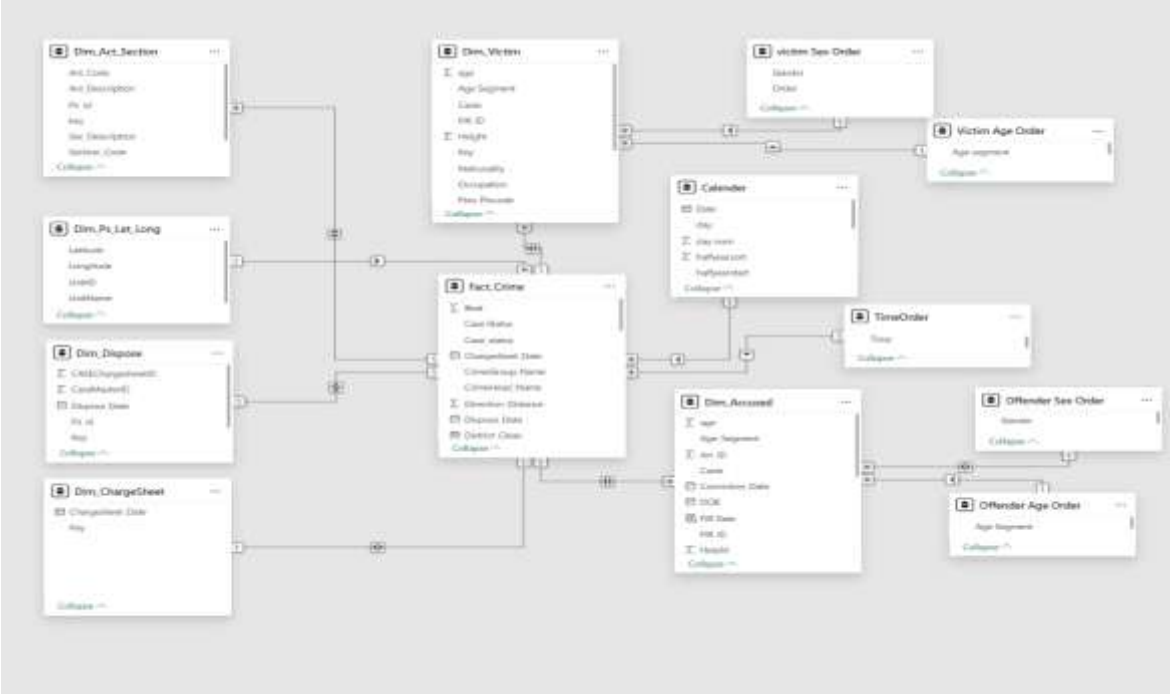
2	Generate Calendar	Created a continuous date range with CALENDAR(startdate, enddate)	Provides a dedicated date dimension required for time intelligence
3	Add Year & Month	Added Year, Month, and Month Name columns	Enables month/year-based aggregation and ensures chronological sorting
4	Add Quarter & Half-Year	Created Quarter, HalfYearStart, HalfYearSort	Allows grouping by larger time buckets for trend analysis
5	Add Week & Day	Created WeekNum, Day, Day Num	Enables weekly trends and weekday/weekend analysis
6	Add Composite Keys	Created Year-Month and sort columns	Supports proper ordering of visuals (e.g., Jan → Feb → Mar, not alphabetical)
7	Use in Model	Connected Calendar with FIR_Date, Chargesheet_Date, and Dispose_Date in fact table	Standardizes multiple date fields into one dimension for consistent analysis
8	Analysis Impact	Used Calendar table in DAX (YoY, MoM, YTD, time differences)	Enables full time-series analysis: trends, seasonality, average processing times

8.Relationships in Your Data Model



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1. **Dim_Accused (Key) → Fact_Crime (Key)**
    a. **Cardinality**: Many-to-One (*:1)
    b. **Cross Filter Direction**: Both
    c. **Purpose :** To analyze crimes based on offender demographics.
    d. Why This Choice: A single crime (FIR) can have multiple accused, but each accused must belong to one FIR → so Many accused → One crime. Both filter is chosen so we can slice either from accused to crime or crime to accused.
2. **Dim_Victim (Key) → Fact_Crime (Key)**
    a. **Cardinality**: Many-to-One (*:1)
    b. **Cross Filter Direction**: Both
    c. **Purpose:** To analyze crimes based on victim demographics
    d. **Why This Choice:** A single crime can have many victims, but each victim belongs to one crime. **Both filter** allows victim-level and FIR-level analysis.
3. **Dim_Act_Section (Key) → Fact_Crime (Key)**
    a. **Cardinality**: Many-to-One (*:1)
    b. **Cross Filter Direction**: Both
    c. **Purpose:** To analyze crimes by law sections.
    d. **Why This Choice:** One crime may involve multiple sections, but one section entry links to only one crime. Both filter allows analysis by law → crime and crime → law.
4. **Dim_ChargeSheet (Key) → Fact_Crime (Key)**
    a. **Cardinality**: One-to-One (1:1)
    b. **Cross Filter Direction**: Both
    c. **Purpose:** To track case progression.
    d. **Why This Choice:** One crime = one chargesheet. If there were multiple chargesheets per crime, it would be 1: but here it's 1:1. Both filter so you can filter from either side.
5. **Dim_Dispose (Key) → Fact_Crime (Key)**
    a. **Cardinality**: One-to-One (1:1)
    b. **Cross Filter Direction**: Both
    c. **Purpose:** To check final case status.
    d. **Why This Choice:** Each case has only one disposal status. That's why 1:1. Both makes it easy to filter either by disposal status or crime.
6. **Dim_Ps_Lat_Long (UnitName) → Fact_Crime (UnitName)**
    a. **Cardinality**: Many-to-One (*:1)
    b. **Cross Filter Direction**: Both
    c. **Purpose:** To map crimes to police stations
    d. **Why This Choice:** A police station handles many crimes, but each crime belongs to one police station → Many-to-One. Both allows slicing from station → crime and vice versa.
7. **Calendar (Date) → Fact_Crime (FIR_Date)**
    a. **Cardinality**: Many-to-One (*:1)
    b. **Cross Filter Direction**: Single
    c. **Purpose:** To support time intelligence.
    d. **Why This Choice:** Many crimes can happen on one date, but one crime can only happen on one date → Many-to-One. Filter direction kept Single to avoid circular filtering with other date fields.

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8. **TimeOrder (Time) → Fact_Crime (Offence Time Period)**
    a. **Cardinality**: Many-to-One (*:1)
    b. **Cross Filter Direction**: Single
    c. **Purpose:** To analyze by time of occurrence.
    d. **Why This Choice:** Many crimes can occur in the same time slot, but one crime has only one time. **Single filter** avoids loops in the model.

9. **Victim Age Order (Age Segment) → Dim_Victim (Age Segment)**
    a. **Cardinality**: Many-to-One (*:1)
    b. **Cross Filter Direction**: Both
    c. **Purpose:** To group victims by age ranges.
    d. **Why This Choice:** Many victims fall in the same age segment, but one victim has only one age segment. Both filter makes it flexible.
10. **Victim Sex Order (Gender) → Dim_Victim (Sex)**
    a. **Cardinality**: Many-to-One (*:1)
    b. **Cross Filter Direction**: Both
    c. **Purpose:** To group victims by gender.
    d. **Why This Choice:** Many victims can be Male/Female/Other, but each victim has only one gender. Both is used for easy drill-down.
11. **Offender Age Order (Age Segment) → Dim_Accused (Age Segment)**
    a. **Cardinality**: Many-to-One (*:1)
    b. **Cross Filter Direction**: Both
    a. **Purpose:** To group accused by age ranges.
    b. **Why This Choice:** Many accused belong to one age segment. Both filter lets us go either way.
12. **Offender Sex Order (Gender) → Dim_Accused (Sex)**
    a. **Cardinality**: Many-to-One (*:1)
    b. **Cross Filter Direction**: Both
    a. **Purpose:** To group accused by gender.
    b. **Why This Choice:** Many accused can share the same gender, but each accused has only one gender. Both filter is used for flexibility.

9.Measures

9.1Reporting & Assign

Measure Name	DAX Formula	Purpose	Logic
Reported Cases	= DISTINCTCOUNT(Fact_Crime[key])	Count all FIRs	Counts distinct case IDs from Fact_Crime.
False Case (B)	= VAR false_cases = CALCULATE(DISTINCTCOUNT(Fact_Crime[Key]) ,Fact_Crime[FIR_Stage] = "Not Investigated / False Case") RETURN	Identify false cases	Filters FIR_Stage and counts false cases.

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	false_cases		
Transferred Case	= VAR transferred_cases = CALCULATE(DISTINCTCOUNT(Fact_Crime[Key]),Fact_Crime[FIR_Stage] = "Transferred") RETURN transferred_cases	Identify transferred cases	Counts cases marked as "Transferred".
Assigned Cases	= ([Reported Cases]-[false case (B)]-[transferred Case])	Valid FIRs	Removes false & transferred from reported.
Assigned Cases Last Year	= CALCULATE([Assigned Cases] ,SAMEPERIODLASTYEAR(Calender[Date]))+0	Compare YOY	Shifts context back 1 year for assigned cases.
Assigned Cases Variance	= VAR __YearSelected = COUNTROWS(VALUES(Calender[year])) VAR __BASELINE_VALUE = [Assigned Cases Last Year] VAR __VALUE_TO_COMPARE = [Assigned Cases] RETURN IF(__BASELINE_VALUE=0 || ISBLANK(__BASELINE_VALUE) || __YearSelected>1 , "N/A", DIVIDE(__VALUE_TO_COMPARE - __BASELINE_VALUE, __BASELINE_VALUE))		ISBLANK(__BASELINE_VALUE)
Assigned Cases Percentage	= VAR TotalCasesAssigned = [Assigned Cases] VAR TotalCasesConstant = [Total Cases (Constant)] VAR unitperc = DIVIDE(TotalCasesAssigned, TotalCasesConstant, 0) RETURN unitperc	% Assigned cases	Compares assigned cases to constant baseline.

9.2Pending

Measure Name	DAX Formula	Purpose	Logic

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Pending Cases (PS)	= VAR active_cases_ps = CALCULATE(DISTINCTCOUNT(Fact_Crime[Key]),Fact_Crime[Case_status] = "Pending@PS") RETURN active_cases_ps	Pending at PS	Filters Case_status = Pending@PS.
Pending Cases (Court)	= VAR active_case_court = CALCULATE(DISTINCTCOUNT(Fact_Crime[key]),Fact_Crime[Case_status] = "Pending@Court") RETURN active_case_court	Pending at Court	Filters Case_status = Pending@Court.
Pending Cases	= ([Pending Cases (Court)]+[Pending Cases (PS)])	Total pending	Adds PS + Court pending.
Pending Cases (%)	= DIVIDE([Pending Cases] , [Reported Cases],0)	Backlog %	Pending ÷ Reported.
Pending Cases Last Year	= CALCULATE([Pending Cases],SAMEPERIODLASTYEAR(Calender[Date]))+0	Compare YOY	Shifts context back 1 year.
Pending Cases YOY %	= VAR CurrentYear_YTD =CALCULATE (TOTALYTD ([Pending Cases], Calender[Date])) VAR PreviousYear_YTD =CALCULATE (TOTALYTD ([Pending Cases], Calender[Date]),DATEADD (Calender[Date], -1, YEAR)) RETURN DIVIDE (CurrentYear_YTD - PreviousYear_YTD, PreviousYear_YTD)	YOY change	Year-over-year % change for pending.

Pending Cases Variance Colour	=
	IF (ISBLANK (
	[Reported cases variance]),
	"#FFFFFF", -- White
	IF (
	[Reported cases variance] > 0,
	"#FF0000", -- Red
	"#008000" -- Green
	))
		Visual aid	Returns Red/Green/White based on variance.

9.3Disposed

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Measure Name	DAX Formula	Purpose	Logic
Disposed Cases (PS)	= CALCULATE(DISTINCTCOUNT(Fact_Crime[Key]),Fact_Crime[Case_status] = "Disposed@PS")	Disposed at PS	Filters Case_status.
Disposed Cases (Court)	= CALCULATE(DISTINCTCOUNT(Fact_Crime[Key]),Fact_Crime[Case_status] = "Disposed@Court")	Disposed at Court	Filters Case_status.
Cases Disposed	= [disposed cases(Court)]+[disposed cases(PS)]	Total disposed	Adds court + PS disposed.
Disposed Rate	= DIVIDE([Cases disposed],[Reported Cases],0)	Clearance rate	% disposed out of reported.
Disposed Rate (Court)	= DIVIDE ([disposed cases(Court)],[Reported Cases],0)	Court disposal rate	Proportion disposed in court.
Disposed Rate (PS)	= DIVIDE ([disposedcases(PS)], [ReportedCases],0)	PS disposal rate	Proportion disposed in PS.
Disposal Cases YOY %	= VAR CurrentYear_YTD =CALCULATE (TOTALYTD ([Cases disposed], Calender[Date])) VAR PreviousYear_YTD =CALCULATE (TOTALYTD ([Cases disposed], Calender[Date]),DATEADD (Calender[Date], -1, YEAR)) RETURN DIVIDE (CurrentYear_YTD - PreviousYear_YTD, PreviousYear_YTD)	YOY change	Year-over-year % change in disposals.

9.4Chargesheet

Measure Name	DAX Formula	Purpose	Logic
Cases Chargesheeted (A)	= VAR tcc = CALCULATE(DISTINCTCOUNT(Fact_Crime[Key]), ISBLANK(Fact_Crime[Chargesheet_Date])= FALSE) RETURN (tcc)+0	Count chargesheeted	Counts cases with Chargesheet_Date.

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Cases Chargesheeted ≤60 Days	= VAR tcc = CALCULATE(DISTINCTCOUNT(Fact_Crime[Key]),ISBLANK(Fact_Crime[Chargesheet_Date]) = FALSE, Fact_Crime[Time Diff (CD - FD)]<=60) RETURN (tcc)+0	Timely chargesheet	Cases completed within 60 days.
Chargesheet Rate v1	= ([Cases Chargesheeted(A)] / [Reported Cases])	Chargesheet rate	Out of reported cases.
Chargesheet Rate ≤60 Days (%)	= DIVIDE([cases Chargesheeted Within 60days],[Cases Chargesheeted(A)],0)	Timely rate	% of chargesheeted cases filed within 60 days.

9.5Time Metrics

Measure Name	DAX Formula	Purpose	Logic
Average Chargesheet Time	= AVERAGE(Fact_Crime[Time Diff (CD - FD)])	Avg duration	Avg days to file chargesheet.
Average Disposible Time	= AVERAGE(Fact_Crime[Time Diff(DD-FD)])	Avg duration	Avg days to dispose.
Average Pending Time (PS)	= AVERAGE(Fact_Crime[Pending Time (PS)])	Pending PS	Avg days cases stuck at PS.
Average Pending Time (Court)	= AVERAGE(Fact_Crime[Pending Time (Court)])	Pending Court	Avg days cases stuck at Court.
Average Pending Time	= VAR ps = [average pending time (PS)] VAR court = [averge pending time (court)] VAR tpt = ps+court RETURN tpt+0	Combined	Adds both PS + Court pending.

9.6YOY, Icons & Colours

Measure Name	DAX Formula	Purpose	Logic
Reported Cases Variance	= VAR CurrentYear_YTD = CALCULATE (	YOY change	Year-over-year % change in reported.

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	TOTALYTD ([Reported Cases], Calender[Date])) VAR PreviousYear_YTD = CALCULATE (TOTALYTD ([Reported Cases], Calender[Date]), DATEADD (Calender[Date], -1, YEAR)) RETURN DIVIDE (CurrentYear_YTD - PreviousYear_YTD, PreviousYear_YTD)		
Reported Cases Icon	= VAR PositiveIcon = "▲" VAR NegativeIcon = "▼" VAR Nochange = "–" VAR __YearSelected = COUNTROWS(VALUES(Calender[year])) VAR __BASELINE_VALUE = CALCULATE(TOTALYTD([Reported Cases], Calender[Date]), DATEADD(Calender[Date], -1, YEAR)) VAR __VALUE_TO_COMPARE = [Reported Cases] VAR yoy = DIVIDE(__VALUE_TO_COMPARE - __BASELINE_VALUE, __BASELINE_VALUE) VAR Result = SWITCH(TRUE(), __BASELINE_VALUE = BLANK() && __VALUE_TO_COMPARE > 0, Nochange, __BASELINE_VALUE = BLANK() && __VALUE_TO_COMPARE = BLANK(), Nochange, __YearSelected > 1, Nochange, yoy > 0, PositiveIcon, yoy < 0, NegativeIcon, Nochange) RETURN Result	Visual trend	Returns arrow icons based on YOY.
Assigned Cases Icon	= var PositiveIcon = "▲" var NegativeIcon = "▼" var Nochange = "–"	Visual trend	Returns arrow icons based on YOY.

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	`VAR __YearSelected = COUNTROWS(VALUES(Calender[year]))`
	
	`VAR __BASELINE_VALUE = [Assigned Cases Last Year]`
	`VAR __VALUE_TO_COMPARE = [Assigned Cases]`
	
	`VAR yoy = DIVIDE(__VALUE_TO_COMPARE - __BASELINE_VALUE, __BASELINE_VALUE)`
	
	`var Result = SWITCH(`
	`  TRUE(),`
	`  __BASELINE_VALUE = BLANK() && __VALUE_TO_COMPARE > 0 ,Nochange,`
	`  __BASELINE_VALUE = BLANK() && __VALUE_TO_COMPARE = BLANK(), Nochange,`
	`  __YearSelected>1, Nochange,`
	`  yoy > 0, PositiveIcon,`
	`  yoy < 0, NegativeIcon,`
	`  Nochange`
	`)`
	
	`RETURN Result`		

Total Cases Variance Colour	`=`
	`IF (`
	`  ISBLANK ( [Reported cases variance] ),`
	`  "#FFFFFF", -- White`
	`  IF (`
	`  [Reported cases variance] > 0,`
	`  "#FF0000", -- Red`
	`  "#008000" -- Green`
	`  )`
	`)`
		Visual formatting	Red/Green/White colours.

Total Cases (Court)	`=`
	`VAR totalcases_court = [Pending Cases (Court)] + [disposed cases(Court)]`
	`RETURN (totalcases_court)+0`
		Visual formatting	Colours for assigned variance.

Total Cases (PS)	`=`
	`VAR totalcases_court = [Pending Cases (PS)] + [disposed cases(PS)]`
	`RETURN (totalcases_court)+0`
		Court load	Total = pending + disposed.

Total Cases (Constant)	`=` `CALCULATE([Assigned Cases], ALL())`	PS load	Total = pending + disposed.

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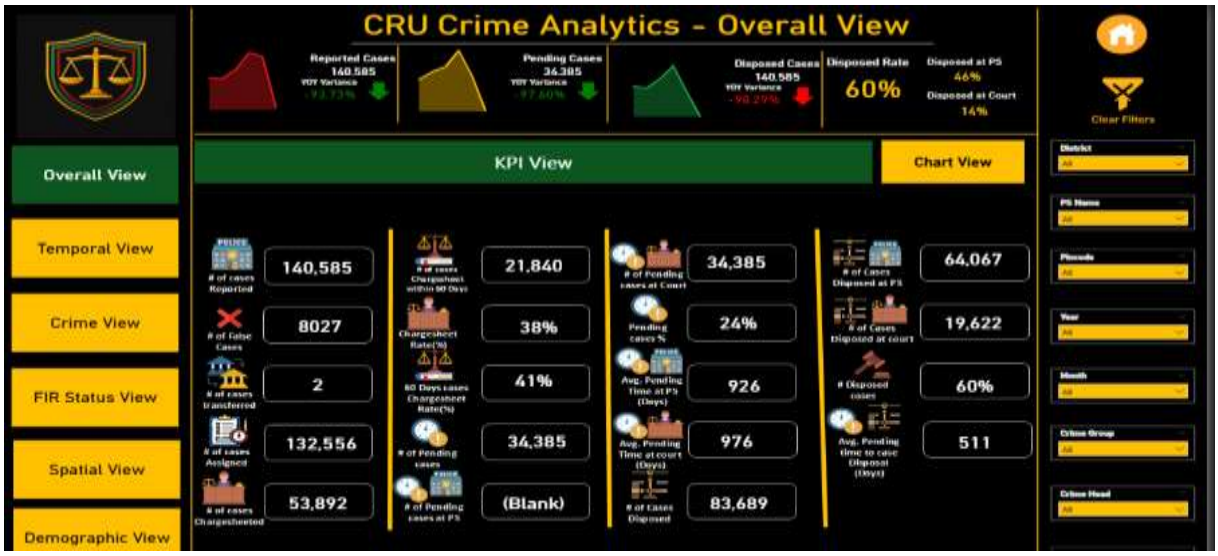
10. Dashboard

10.1 CRU Crime Analytics-Overall View



10.1.1 Reporting & Assignment KPIs

KPI Name	Measure(s) Used	Purpose	Insight	Problems / Limitations
# of Cases Reported	[Reported Cases], [Reported Cases Variance]	Show total FIRs & YoY change	Reported cases declined sharply (-93.7% YoY).	Drop suggests data quality/coverage issue.
# of False Cases	[False Case (B)]	Track false or non-investigated cases	False cases reduced (-85.6% YoY).	Year-on-year tagging may be inconsistent.
# of Cases Transferred	[Transferred Case]	Count transferred cases	Transfers nearly disappeared (-99.5% YoY).	Needs verification at district level.
# of Cases Assigned	[Assigned Cases]	Show valid FIRs after exclusions	Case assignments dropped with reported cases.	Assignment criteria may have changed.

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10.1.2 Chargesheet & Investigation KPIs

KPI Name	Measure(s) Used	Purpose	Insight	Problems / Limitations
# of Cases Chargesheeted	[Cases Chargesheeted (A)]	Count cases moved to chargesheet stage	Fewer cases moved to chargesheet stage.	Risk of backlog if inflow > disposal.
# of Cases Chargesheeted ≤60 Days	[Cases Chargesheeted Within 60days]	Track timeliness of filing	Timely chargesheeting reduced sharply (-88% YoY).	Identify PS with delays >60 days.
Chargesheet Rate (%)	[Chargesheet Rate v1]	Share of reported cases → chargesheet	Chargesheet rate down to 38%.	Efficiency slipping; compare vs benchmarks.
60-Day Chargesheet Rate (%)	[Chargesheet Rate within 60days (%)]	Proportion chargesheeted ≤60 days	Timely rate dropped to 41%.	Needs breakdown by crime type.

10.1.3 Pending Cases & Delay KPIs

KPI Name	Measure(s) Used	Purpose	Insight	Problems / Limitations
# of Pending Cases	[Pending Cases], [Pending Cases YOY %]	Total backlog cases	Pending backlog reduced sharply (-91% YoY).	May reflect reclassification or data issue.
Pending Cases at Court	[Pending Cases (Court)]	Court backlog	Court pendency reduced to 34K.	Drop unusually steep — validate with court records.
Pending Cases at PS	[Pending Cases (PS)]	PS backlog	PS pendency reduced to 22K.	Possible incomplete PS updates.
Pending Cases %	[Pending Cases (%)]	% of cases pending	Pendency share fell to 40%.	Check denominat

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				or consistency.

10.1.4 Disposal KPIs

KPI Name	Measure(s) Used	Purpose	Insight	Problems / Limitations
# of Cases Disposed	[Cases Disposed], [Disposal Cases YOY %]	Total disposals	Steep decline in disposals (-98% YoY).	Indicates systemic slowdown.
Cases Disposed at PS	[Disposed Cases (PS)]	Disposals at PS	PS disposals dropped to 64K (-96%).	Negative efficiency trend.
Cases Disposed at Court	[Disposed Cases (Court)]	Disposals at Court	Court disposals dropped to 19K.	Must check per-district court workloads.
Disposed Rate (%)	[Disposed Rate]	Clearance % of reported cases	Disposal rate fell to 60%.	Efficiency reduced.

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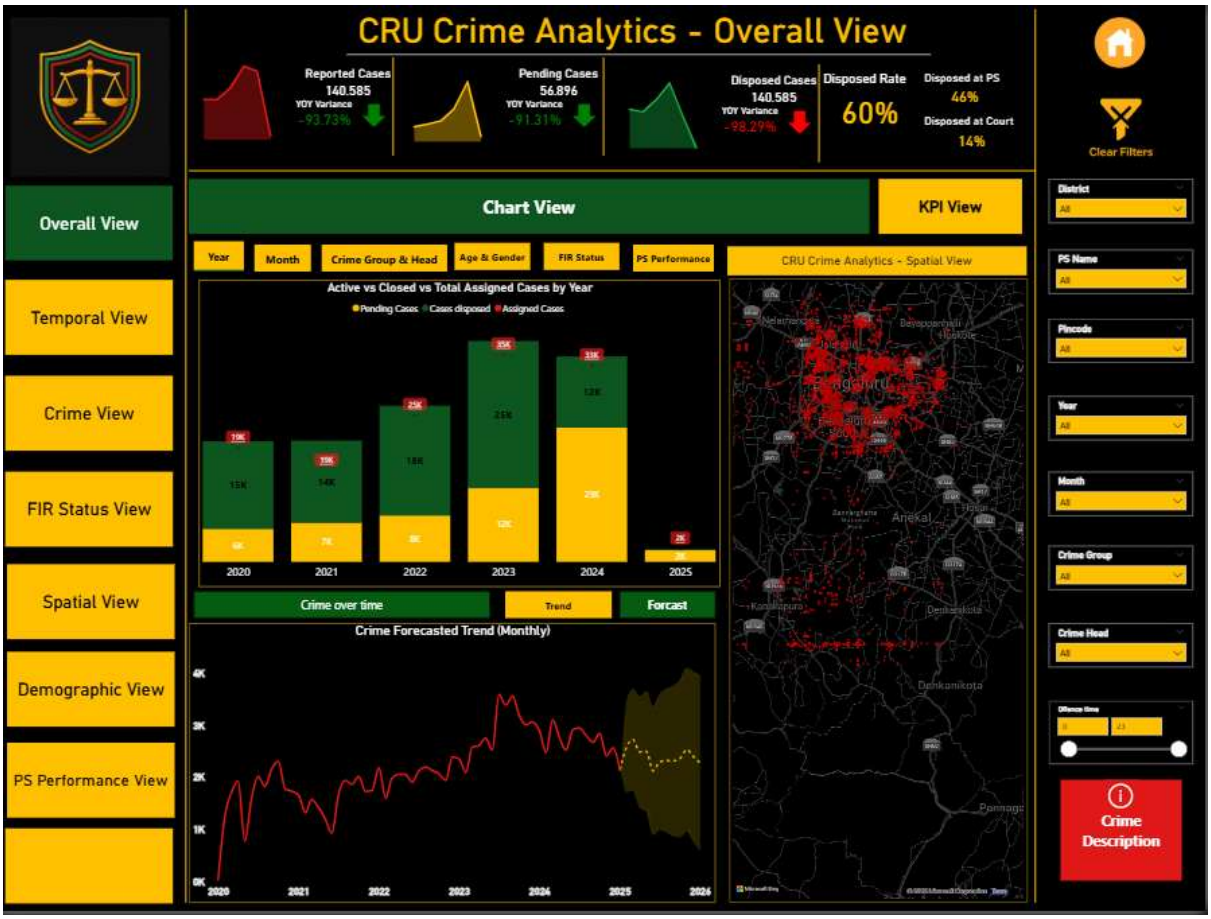

Button	Bookmark Triggered	Visuals Shown	Visuals Hidden	Notes
Month Trends	MonthTrend_BM	Crimes by Month (clustered bar chart)	Year Trends, Crime Group Trends, Age & Gender Slicer, FIR Status Chart, PS Performance, Forecast, Trends	Shows seasonal variation
Year Trends	YearTrend_BM	Crimes by Year (line/column chart)	Others	Highlights yearly growth/decline
Crime Group Trends	CrimeGroupTrend_BM	Heinous vs Non-Heinous Crimes (bar chart)	Others	Connects with Crime View page
Age & Gender Slicer	AgeGender_BM	Victim demographics (Age, Gender)	Others	Demographic filter applied
FIR Status Chart	FIRStatus_BM	Pending vs Disposed FIRs (bar chart)	Others	Quick FIR snapshot
PS Performance	PSPerf_BM	Pending vs Disposed by Police Station	Others	Shortcut to PS page
Forecast	Forecast_BM	Crime Forecast 2026	Others	Based on 2020–2024, excludes 2025

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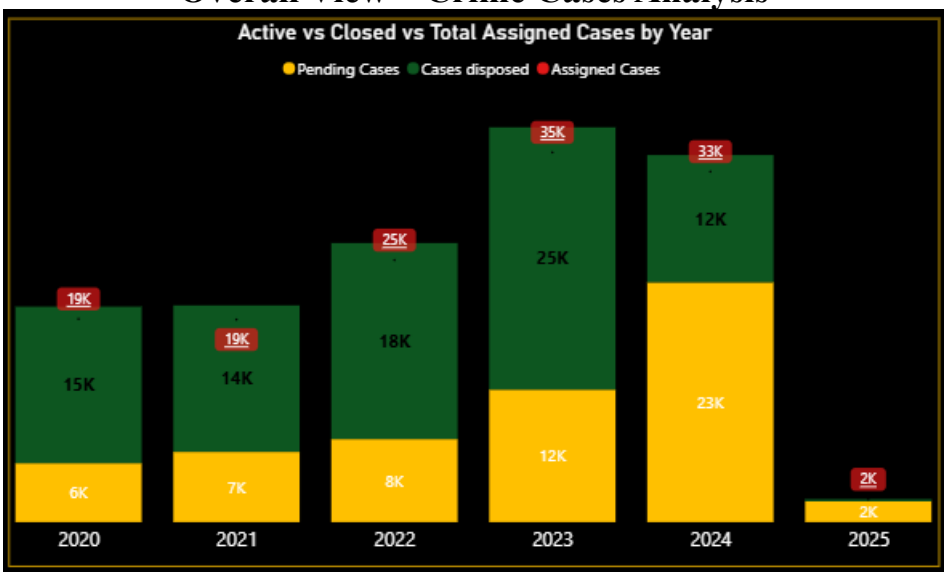
Trends	Trends_BM	General crime trend line (default view)	All other variations	Default state when page loads

Overall View – Crime Cases Analysis



Active vs Closed vs Total Assigned by Year

Aspect	Details
Visual Type	Line & Stacked Column Chart
Data Fields	**X-Axis:** Year Column Y-Axis: Pending Cases, Cases Disposed **Line Y-Axis:** Assigned Cases
Measure(s) Used	[Assigned Cases], [Pending Cases], [Cases Disposed]
Purpose	Show year-wise lifecycle trend of Assigned, Pending, and Disposed cases.
Insights	- Assigned cases peaked in **2023 (35K)** before falling in 2024–2025. - Pending cases also high in 2023, backlog accumulation. - Sharp decline in 2025 (only 2K cases), possible data entry cutoff.

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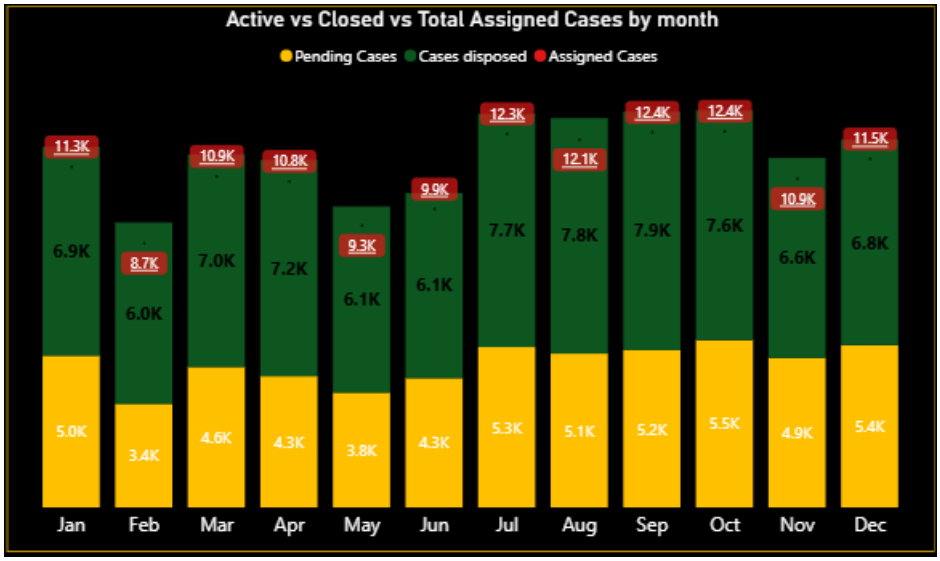

Active vs Closed vs Total Assigned Cases – Monthly Trend

Aspect	Details
Visual Type	Line & Stacked Column Chart
Data Fields	**X-Axis:** Month Column Y-Axis: Pending Cases, Cases Disposed **Line Y-Axis:** Assigned Cases
Measure(s) Used	[Assigned Cases], [Pending Cases], [Cases Disposed]
Purpose	Track monthly distribution of assigned, pending, and disposed cases to highlight seasonal patterns and backlog behavior.
Insights	- Assigned cases peak in **July–October (~12K)**. - Pending cases also rise in these months → backlog accumulation. - Disposals steady at ~6K–7K per month. - Jan–Feb show relatively fewer cases.
Problems / Predicted Reasons	- Seasonal spikes July–Oct due to festivals/holidays → more thefts, disputes, social crimes. - Police & courts overloaded mid-year → slower disposals → pending rises. - Lower Jan–Feb cases may be under-reporting or faster closures. - Possible backlog clearance drives at year-end reduce pending cases.

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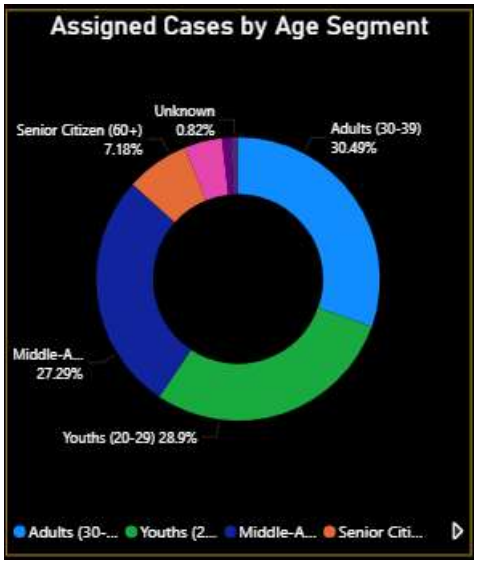

Assigned Cases by Age Segment

Aspect	Details
Visual Type	Donut Chart
Data Fields	**Legend:** Age Segment **Values:** Assigned Cases
Measure(s) Used	[Assigned Cases]
Purpose	Identify which age groups are most affected in assigned crime cases, highlighting vulnerable segments of the population.
Insights	• Adults (30–39) = **30.5%** (largest share). • Youth (20–29) = **28.9%** (closely follows adults). • Middle-aged (40–59) = **27.3%**. • Senior citizens (60+) = **7.2%**. • Children & Unknown = very small shares.
Problems / Predicted Reasons	• Adults & youths more socially/economically active → greater exposure to assaults, fraud, workplace conflicts. • Middle-aged victims linked to property disputes & scams. • Senior citizens, though fewer, targeted in fraud/neglect. • High share of youth victims signals safety concerns for students/job-seekers. • Should cross-check with crime categories & gender for deeper insights.

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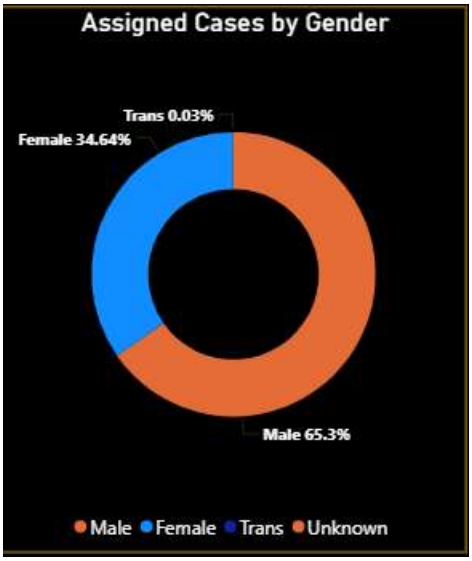

Assigned Cases by Gender

Aspect	Details
Visual Type	Donut Chart
Data Fields	**Legend:** Gender **Values:** Assigned Cases
Measure(s) Used	[Assigned Cases]
Purpose	Understand the gender distribution of victims in assigned crime cases to highlight which gender groups are more impacted.
Insights	• Male victims = **65.3%** (majority). • Female victims = **34.6%**. • Transgender victims = **0.03%** (low but reported). • Unknown = negligible.
Problems / Predicted Reasons	• **Males** more exposed in outdoor/public spaces (workplace, travel, traffic) → higher theft/assault risk. • **Females** more linked to domestic violence, harassment, and gender-specific crimes. • **Transgender victims** low due to under-reporting or smaller population share, though awareness is improving. • Policy concern: Youth + Adults (20–39 yrs) overlap with male dominance → working-age males most vulnerable. • Female share still significant → signals gender-based safety risks.

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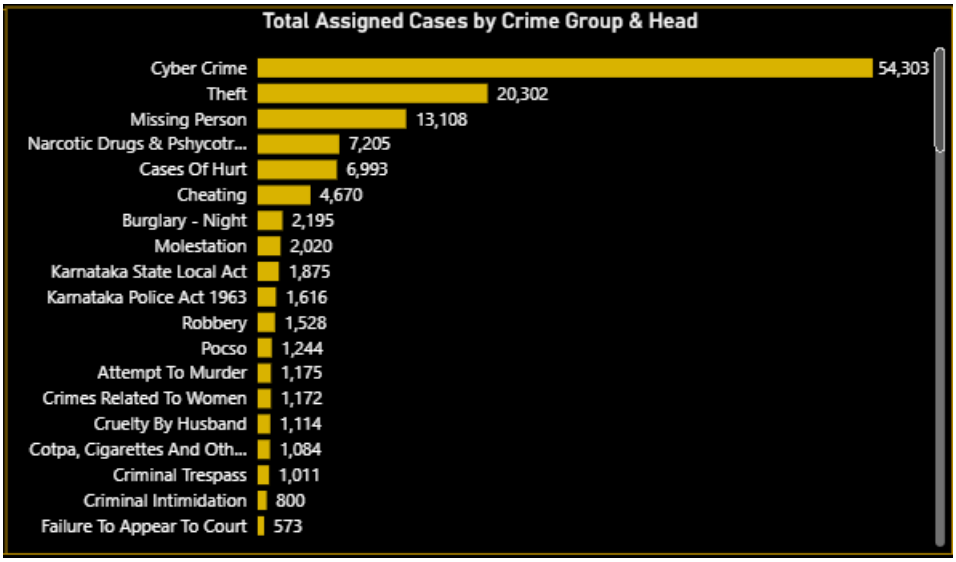

Total Assigned Cases by Crime Group & Head

Aspect	Details
Visual Type	Clustered Bar Chart
Data Fields	**X-Axis:** [Assigned Cases] **Y-Axis:** Crime Group, Crime Head, Act Description, Sec Description
Measure(s) Used	[Assigned Cases]
Purpose	Show distribution of assigned cases across different crime groups and heads to identify the most prevalent categories.
Insights	• Cyber Crime = 54,303 (dominant). • Theft = 20,302; Missing Person = 13,108. • Narcotics & Psychotropic Cases = 7,205. • Gender & Domestic Crimes (Molestation = 2,020; Cruelty by Husband = 1,114). • Violent Crimes (Robbery, Attempt to Murder, Murder) lower but significant. • Other Crimes (Cheating, Trespass, Intimidation) moderate.
Problems / Predicted Reasons	• Cyber Crime: Rise due to digitization, weak cyber hygiene, phishing scams. • Theft & Missing Persons: Socio-economic issues, migration, trafficking, family disputes. • Narcotics: Drug abuse among youth → need for stronger enforcement. • Gender/Domestic Crimes: Socio-cultural norms, under-reporting, lack of awareness. • Violent Crimes: Lower than cyber but still highlight public safety risks. • Other Crimes: Everyday law & order issues; requires routine policing.

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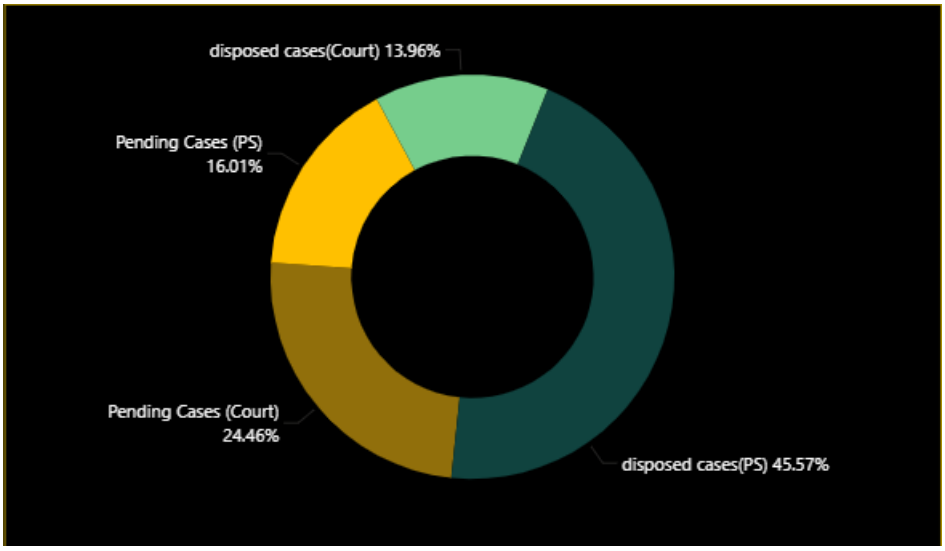

FIR Status – Pending vs Disposed Cases

Aspect	Details
Visual Type	Donut Chart
Data Fields	**Legend/Category:** Pending Cases (PS), Pending Cases (Court), Disposed Cases (PS), Disposed Cases (Court)
Measures Used	[Disposed Cases (Court)], [Disposed Cases (PS)], [Pending Cases (Court)], [Pending Cases (PS)]
Purpose	Track case disposal efficiency at Police Station (PS) and Court levels, and identify where delays are happening.
Insights	• **Disposed Cases (PS): 45.6%** → Nearly half resolved at PS level (good sign). • **Pending Cases (Court): 24.5%** → Largest backlog. • **Pending Cases (PS): 16%** → Moderate, linked to investigation delays. • **Disposed Cases (Court): 14%** → Very low compared to pending court cases.
Problems / Predicted Reasons	• **Court Backlog:** Judicial delays, shortage of judges, adjournments, procedural complexity. • **PS Backlog:** Incomplete investigations, weak evidence, staff shortages. • **Court Disposal Low:** Cases pile up in trial stage; courts less efficient than PS.
What Needs to Improve	• Speed up judicial process: fast-track courts, more judges, digital case systems. • Better coordination between PS & courts for timely evidence/chargesheets. • Monitor high-pendency courts, allocate more resources. • Set disposal targets (e.g., ≥70% within 12 months).

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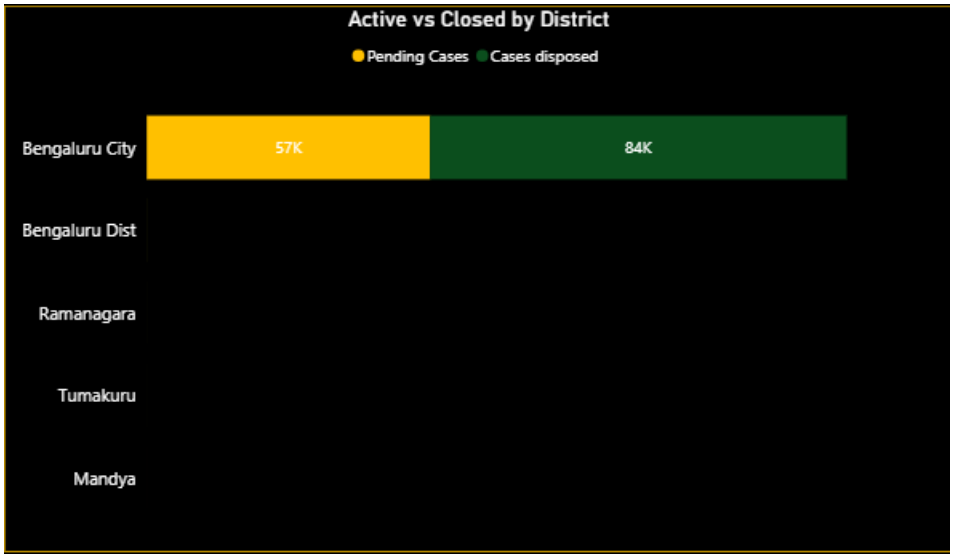

Active vs Closed by District

Aspect	Details
Visual Type	Stacked Bar Chart
Data Fields	**Y-Axis:** District_Name **X-Axis:** Pending Cases, Cases Disposed
Measures Used	[Cases disposed], [Pending Cases]
Purpose	Compare pending vs disposed FIR cases across districts, identify backlog hotspots, and guide resource allocation.
Insights	• **Bengaluru City:** 57K pending vs 84K disposed → strong disposal but backlog still very high. • **Other districts (Bengaluru Dist, Ramanagara, Tumakuru, Mandya):** lower volumes but significant pending relative to disposed. • Bengaluru City dominates both pending & disposed, reflecting higher caseload and efficiency compared to smaller districts.
Problems / Predicted Reasons	• **Bengaluru City:** Huge absolute backlog despite strong performance, likely due to massive inflow of cases. • **Smaller districts (Tumakuru, Mandya):** Low disposal efficiency, possibly staff shortage, investigation bottlenecks, or judicial delays.
What Needs to Improve	• **Bengaluru City:** Add more judicial/police staff, introduce backlog clearance drives. • **Smaller districts:** Conduct audits to identify bottlenecks; deploy additional resources. • Use **technology-driven tracking systems** to monitor pending cases.

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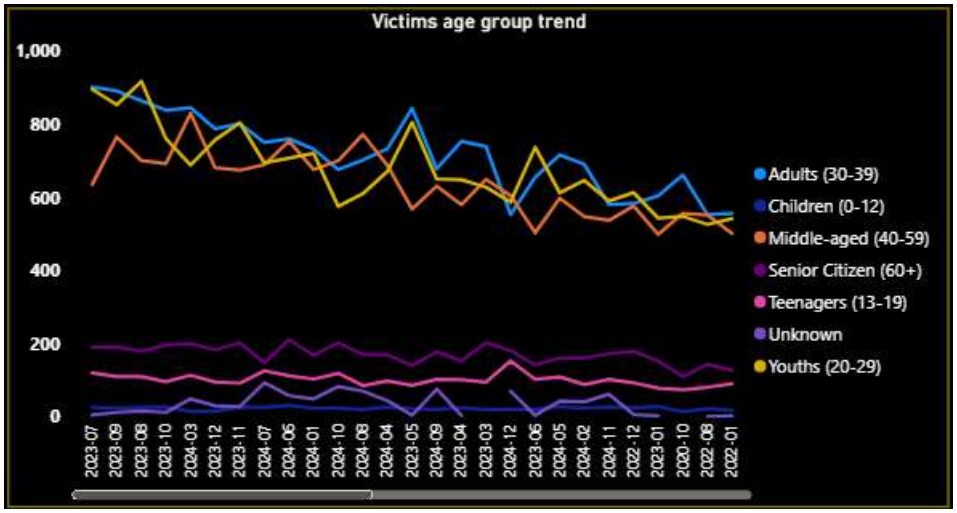
	• Review YoY trend to check if backlog is reducing or worsening.



Victims Age Group Trend (Monthly Analysis)

Aspect	Details
Visual Type	Line Chart
Data Fields	**X-Axis:** Year-Month **Y-Axis:** [Assigned Cases] **Legend:** Victim Age Segment
Measures Used	[Assigned Cases]
Purpose	Identify vulnerable age groups, track monthly fluctuations, and detect seasonal victimization patterns to guide targeted interventions.
Insights	• Adults (30–39) and Children (0–12) record the highest cases → most vulnerable groups. • Youths (20–29): strong presence but slightly lower. • Teenagers (13–19) and Seniors (60+): consistently lower, more stable. • Strong month-to-month fluctuations → seasonal/social influences (festivals, migration, school/college cycles). • Recent convergence: Adults & Children trending closer to Youth levels → may reflect policy interventions or reporting changes.
Problems / Predicted Reasons	• Children (0–12): High vulnerability → trafficking, abuse, neglect, weak child safety enforcement. • Adults (30–39): High exposure due to workplace stress, financial fraud, domestic violence. • Volatile trends: Preventive mechanisms not consistent.

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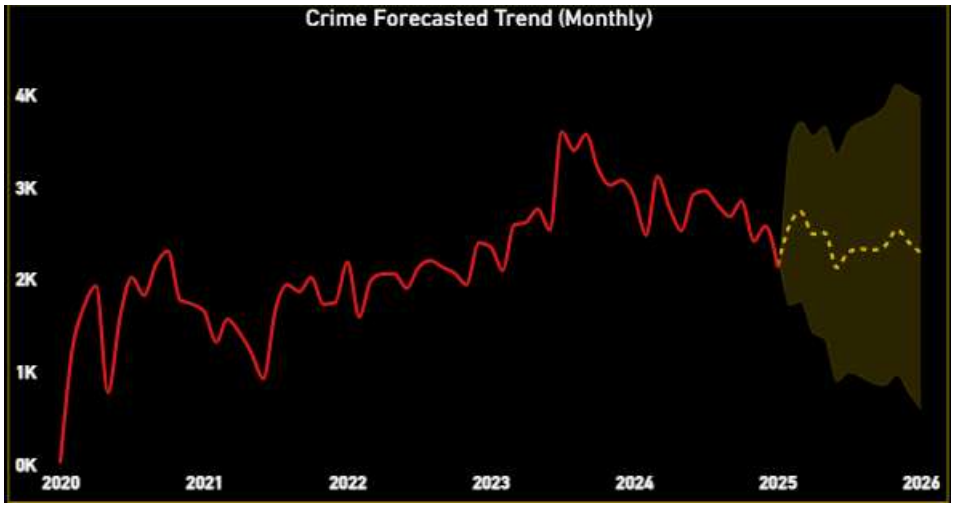
What Needs to Improve	• Strengthen child protection systems (school monitoring, helplines, awareness drives). • Targeted support for adults (30–39) → workplace safety, domestic violence support, financial crime prevention. • Seasonal crime prediction: allocate resources during peak months. • Link with Crime Type Analysis to find which crimes affect children vs adults. • Community awareness programs tailored to children, parents, and working-age adults.



Crime Forecasted Trend (Monthly)

Aspect	Details
Visual Type	Line Chart
Data Fields	**X-Axis:** YearMonthStart, Year, Quarter, Month, Day **Y-Axis:** Assigned Cases
Measures Used	[Assigned Cases]
Formatting & Design	Forecast length: 12 months Confidence Interval: 95% Confidence Shade: Yellow Forecast Line: Yellow dotted Historical Trend: Red solid Upper Bound = Worst-case Lower Bound = Best-case
Purpose	To project future monthly crime cases with confidence ranges, helping stakeholders prepare for both most-likely and extreme-case scenarios.
Insights	• Historical (2020–2024): Upward trend peaking >3,500 cases/month. • Forecast (2025–2026): Stabilization expected, but still high. • Q2 2025 Example: Avg. forecast = 2,485 cases/month; • Upper bound = 3,551; Lower bound = 1,420.

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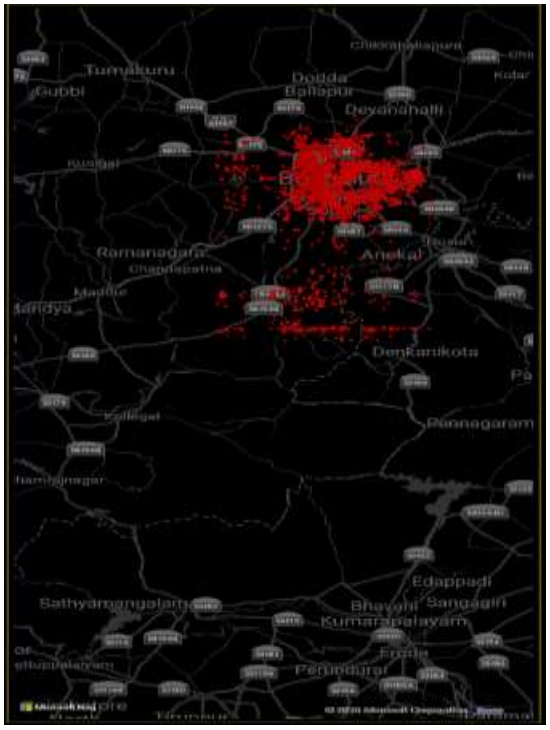
	• Wide confidence band → uncertainty, requiring cautious planning.
Problems / Predicted Risks	• Police workload to remain consistently high. • Big swing in forecasts (±1,000+) means unpredictability in monthly caseload.
Recommendations	• Allocate resources based on upper bound to avoid under-preparation. • Strengthen crime prevention programs to move actuals closer to lower bound. • Continuously compare actual vs forecast to fine-tune model accuracy.



CRU Crime Analytics – Spatial View

Aspect	Details
Visual Type	Map
Data Fields	**Latitude:** lat_Final **Longitude:** Long_Final **Bubble Size:** Assigned Cases **Tooltip:** UnitName
Measures Used	[Assigned Cases]
Purpose	To show spatial distribution of crime incidents across the region, identify hotspots, and compare intensity between urban and rural police jurisdictions.
Insights	• Bengaluru & surrounding urban centers show the highest case density.

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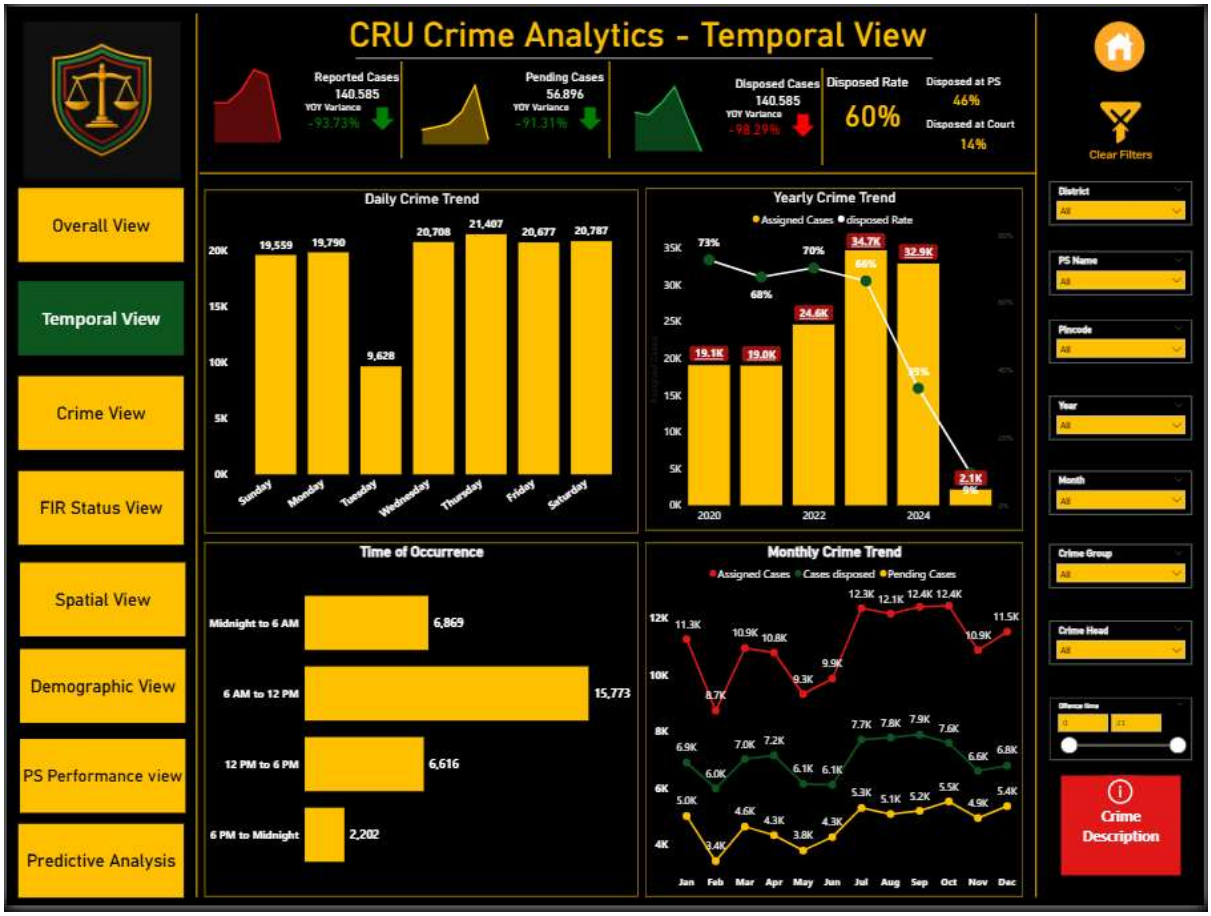
	• Peripheral towns (Tumakuru, Doddaballapur, Anekal) have smaller case volumes. • Heat concentration proves urban centers are more crime-heavy. • Time-slider (0–23 hrs) adds an extra layer of analysis (day vs night crime patterns).
Problems / Predicted Risks	• Urban police stations are overloaded with high caseloads. • Rural areas may be under-reported but still face critical crime types. • Resource imbalance between high-density vs low-density regions.
Recommendations	• Deploy more manpower & resources in urban hotspots. • Use time-of-day trends to optimize police patrol schedules. • Enhance coordination across districts for inter-jurisdiction crimes. • Launch crime prevention & awareness drives in high-density areas like Bengaluru.



Daily Crime Trends (Weekday-wise Assigned Cases)

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Aspect	Details
Visual Type	Stacked Column Chart
Data Fields	**X-Axis:** Day_Name **Y-Axis**: [Assigned Cases]
Measures Used	`[Assigned Cases]`
Purpose	• Show workload distribution of assigned cases across weekdays. • Identify peak days for crime assignment. • Support resource planning and operational deployment.
Insights	• Thursday highest workload (21,407 cases). • Saturday also high (20,487 cases). • Tuesday lowest (9,628 cases). • Other weekdays: moderate range (10K–18K). • Pattern: mid-to-late week (Thu–Sat) busier vs early week (Mon–Tue quieter).
Problems / Predicted Reasons	• Thursday peak → weekly review cycles, backlog pushed midweek. • Saturday spike → weekend crimes (disturbances, theft, traffic) + pre-weekend assignments. • Tuesday dip → fewer fresh cases post-weekend; backlog clears on Mon/Wed. • Imbalance → resource strain on Thu–Sat, underutilization on Mon–Tue.
Recommendations	• Spread case assignment evenly across weekdays. • Break down crime types per weekday to confirm cause of peaks. • Align manpower & patrol deployment with Thu–Sat peaks. • Use automation to reduce clustering. • Monitor monthly/seasonal patterns to check if weekday imbalance is consistent.

Yearly Crime Trends – Assigned Cases vs Disposed Rate

Aspect	Details
Visual Type	Line & Clustered Column Chart
Data Fields	**X-Axis:** Year **Column (Y1):** [Assigned Cases] **Line (Y2):** [Disposed Rate]
Measures Used	[Assigned Cases], [Disposed Rate]
Purpose	Compare yearly crime assignment volumes with disposal efficiency, to evaluate whether rising caseloads impact case clearance.
Insights	• 2023: Highest assignment (34.7K) but disposal only 66%. • 2020: Best disposal efficiency (73%) despite moderate caseload (19.1K).

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	• 2021–22: Steady performance (~19K–24K cases, 68–70% disposal). • 2024: Assignment spiked to 32.9K but disposal collapsed to 35%. • 2025 (YTD): Very low assignment (2.1K) but alarming disposal efficiency (9%) — possibly incomplete data. • Evidence: Higher caseload years → disposal efficiency falls sharply.
Problems / Predicted Reasons	• 2024 Backlog → spike in cases overburdened system. • 2025 Low Disposal → partial year, backlog still unresolved. • Volume vs Efficiency mismatch → 2020 & 2022 efficient despite moderate volume, so resource/process bottlenecks are also factors. • Lack of breakdown by crime type → complex crimes may skew disposal rates.
Recommendations	• Investigate 2024 backlog drivers (policy change, staff shortage, rise in cybercrime/fraud). • Benchmark best years (2020, 2022) → replicate successful strategies. • Add drill-through by crime group for deeper efficiency insights. • Implement workload balancing in high-volume years.

Crime Trends by Time of Occurrence – Assigned Cases

Aspect	Details
Visual Type	Stacked Bar Chart
Data Fields	**Y-Axis:** Time Range **X-Axis:** Assigned Cases **Legend:** Unit Name (Police Unit/Department)
Measures Used	[Assigned Cases]
Purpose	Analyze case assignment trends by time of day to identify peak hours, workload distribution, and resource needs across units.
Insights	• Morning (6AM–12PM): Highest case load (15,773) → majority of crimes reported/assigned in morning. • Midnight–6AM: Second highest (6,869) → indicates significant late-night incidents. • Afternoon (12PM–6PM): Moderate (6,616). • Evening/Night (6PM–Midnight): Lowest (2,202) → suggests underreporting or reporting delays. • Trend: Sharp morning peak, dip in afternoon, very low evening assignment.
Problems / Predicted Reasons	• Morning Peak → delayed reporting of night crimes. • Midnight–6AM High → late-night thefts, assaults, alcohol-related incidents.

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	• Low Evening (6PM–Midnight) → underreporting due to fear, lack of access, or preference to report next morning. • Reporting bias → chart shows assignment time, not exact occurrence time. • Crime-type differences hidden in aggregate (e.g., cybercrime may not follow same pattern).
Recommendations	• Increase late-night patrols (12AM–6AM) to match crime load. • Strengthen morning intake (6AM–12PM) by deploying more station staff. • Awareness drives to encourage evening reporting instead of delaying until morning. • Add crime-type breakdown by time to reveal category-specific peaks. • Use predictive staffing models → align manpower with reporting spikes.

Monthly Crime Trends

Aspect	Details
Visual Type	Line Chart
Data Fields	**X-Axis:** Month (Jan–Dec) **Y-Axis:** Assigned Cases, Disposed Cases, Pending Cases
Measures Used	[Assigned Cases], [Cases disposed], [Pending Cases]
Purpose	Track month-by-month variation in case assignment, disposal, and pending backlog. Helps uncover seasonal patterns and check if enforcement is keeping up with caseloads.
Insights	• Assigned Cases: Peaks in Oct (12.4K), Jul (12.3K), Sep (12.3K) → seasonal surges. Lowest in Feb (8.7K). • Disposed Cases: Range 6.1K–7.9K, with best clearance Jul–Sep (>7.5K). • Pending Cases: Low in Feb (3.4K), but rises sharply in Oct–Dec (>5K). • Biggest Gap: Oct → 12.4K assigned vs 7.6K disposed → backlog 5.4K (worst month for system stress).
Problems / Predicted Reasons	• Backlog surge (Oct–Dec): Likely end-of-year crime/reporting spike + limited disposal bandwidth. • Feb dip: Seasonal reduction (short month, festivals, lower outdoor activity). • Disposals not scaling with assignments → structural bottlenecks in investigation or court process. • Sustained Pending (5K+) indicates systemic delays beyond seasonal fluctuation.
Recommendations	• Boost staffing/resources during Jul–Oct high-load months.

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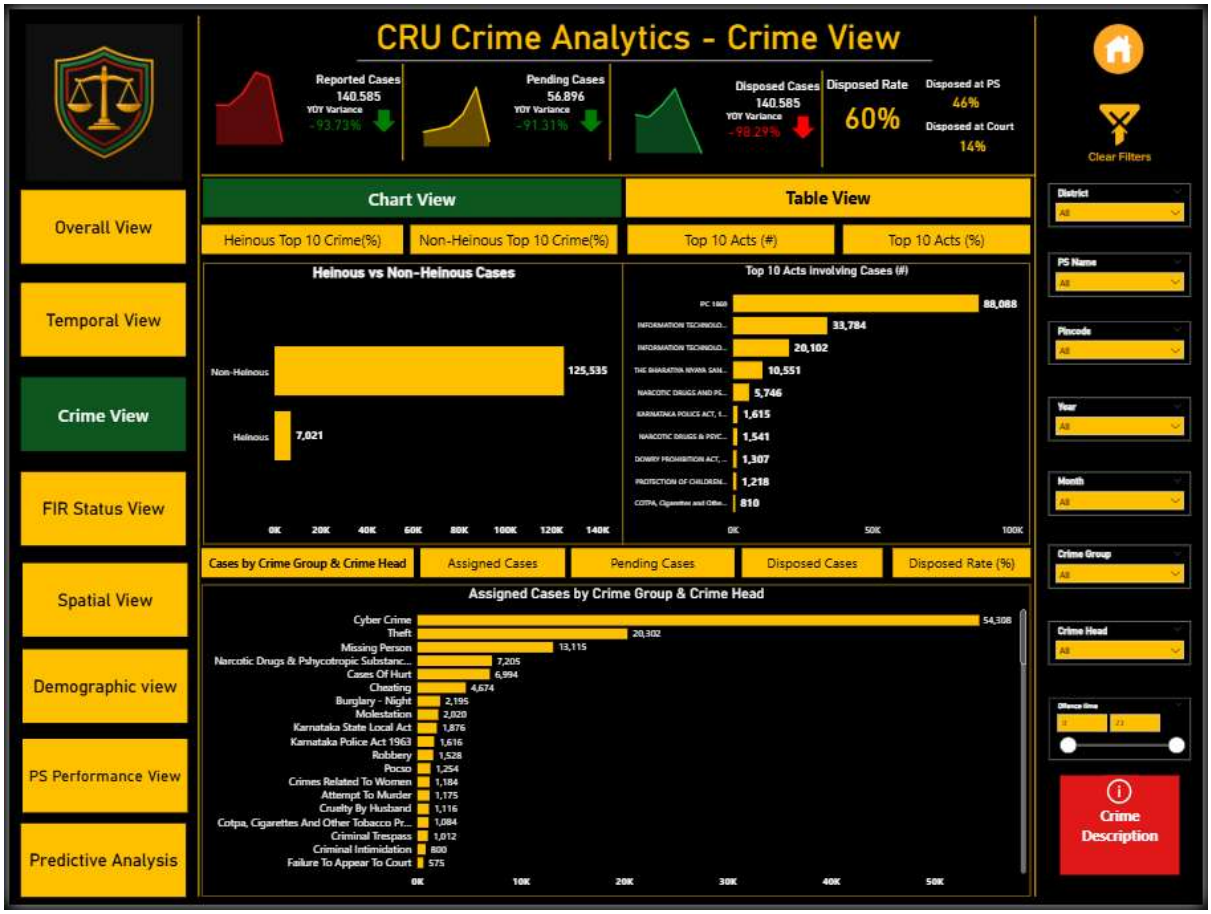
	• Conduct backlog audits for Oct–Dec → courts, police, or evidence bottlenecks? • Compare YoY monthly trends → identify recurring seasonal spikes. • Use Power BI Forecasting to anticipate pending load in upcoming months.



Crime View (Chart View)

Button	Bookmark Triggered	Visuals Shown	Visuals Hidden	Notes
Heinous Top 10 Crimes	HeinousTop10_BM	Top 10 Heinous Crimes (bar chart)	Non-Heinous Top 10, Top 10 Acts, Group-level charts	Focus on severe crimes
Non-Heinous Top 10 Crimes	NonHeinousTop10_BM	Top 10 Non-Heinous Crimes (bar chart)	Heinous Top 10, Top 10 Acts, Group-level charts	Highlights frequent but less severe crimes
Top 10 Acts (#)	Top10ActCount_BM	Top 10 Acts by number of cases	Heinous/Non-Heinous, Top 10 Act (%), Group-level charts	Raw case count

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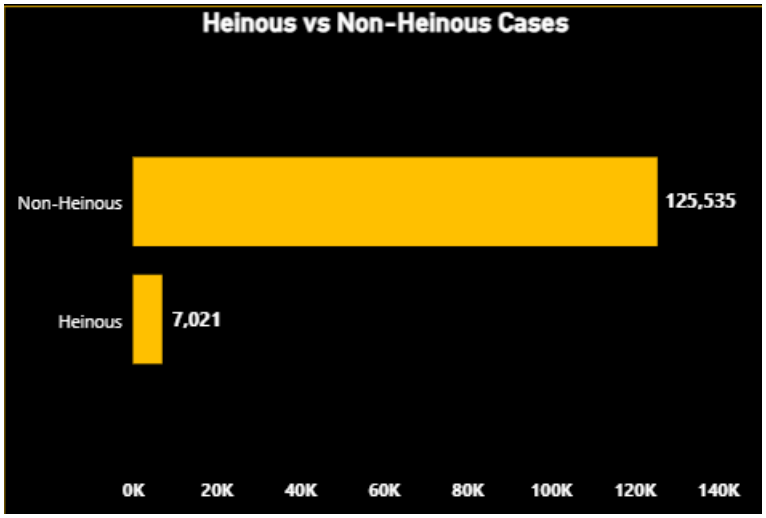
Top 10 Acts (%)	Top10ActPercent_BM	Top 10 Acts by % share	Heinous/Non-Heinous, Top 10 Act (#), Group-level charts	Shows contribution to total
Cases by Crime Group & Category	CasesByGroupCat_BM	Assigned/Disposed/Pending by group & category	All other bookmarks	Group-level split
Assign Cases by Crime Group	AssignCasesGroup_BM	Assigned cases by crime group	Others	Workload comparison
Pending Cases by Crime Group	PendingCasesGroup_BM	Pending cases by crime group	Others	Shows backlog
Disposed Cases by Crime Group	DisposedCasesGroup_BM	Disposed cases by crime group	Others	Efficiency indicator
Disposed Cases (%) by Crime Group	DisposedRateGroup_BM	Disposal % by crime group	Others	KPI for efficiency



Heinous vs Non-Heinous Cases

Aspect	Details
Visual Type	Clustered Bar Chart
Data Fields	**X-Axis:** Assigned Cases (No. of FIRs) **Y-Axis:** FIR Type (Heinous / Non-Heinous)
Measures Used	[Assigned Cases]
Purpose	Compare distribution between heinous and non-heinous crimes, show dominance of non-heinous in total reporting, and provide a drill-down base for age, crime type, and demographics.
Insights	• Non-Heinous Cases: 125,535 (≈94% of total). • Heinous Cases: 7,021 (≈6%).

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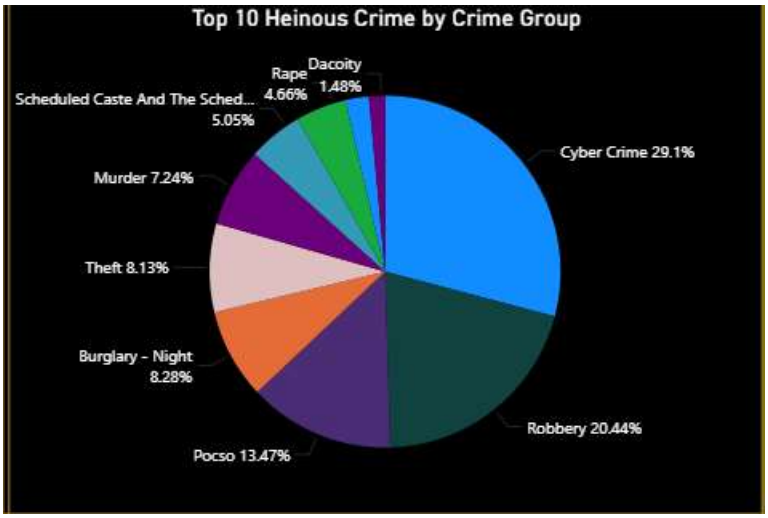
	• Non-heinous dominates overall, but heinous cases demand higher effort despite lower volume. • Localized filters (district, crime group) can change proportions.
Problems / Predicted Reasons	• Dilution of focus: Huge non-heinous load (petty theft, disputes, cyber fraud) may reduce attention on heinous crimes. • Heinous underreporting: Fear, stigma, or inefficiency may hide true volume. • Investigation challenge: Heinous crimes require deeper, time-intensive investigation vs. petty cases.
Recommendations	• Create separate task forces for heinous vs non-heinous to balance workloads. • Fast-track courts & special units for heinous cases. • Apply predictive analytics to detect rising non-heinous crime trends (e.g., cybercrime, theft hotspots). • Run awareness campaigns to improve reporting of heinous crimes.



Top 10 Heinous Crimes

Aspect	Details
Visual Type	Pie Chart (distribution view)
Data Fields	**Legend:** CrimeGroup_Name **Values:** Assigned Cases

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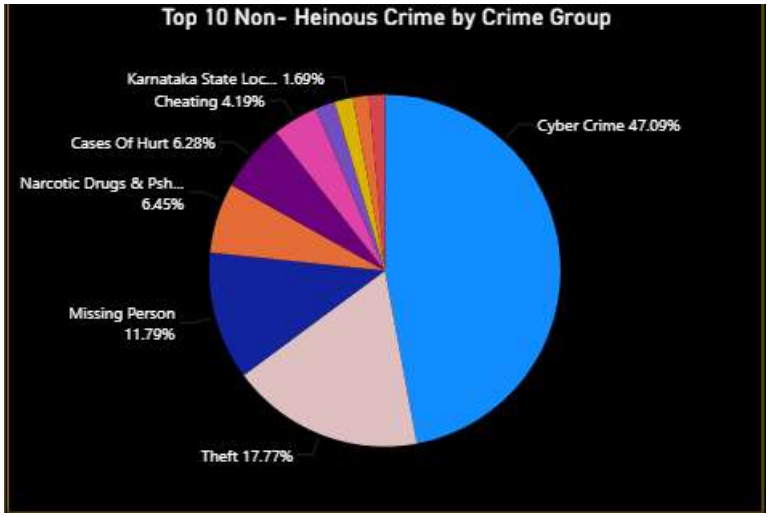
	Filters: FIR Type = Heinous CrimeGroup_Name 10 by Assigned Cases
Measures Used	[Assigned Cases]
Purpose	Identify which heinous crimes dominate reported cases, compare distribution across top crime groups, and highlight focus areas for policing.
Insights	• Cyber Crime: 29.1% (largest share). • Robbery: 20.4% (second-highest). • POCSO: 13.5% (serious child-related crimes). • Burglary at Night (8.3%) & Theft (8.1%) still notable. • Murder: 7.2%. • Top 3 (Cyber + Robbery + POCSO) = ~63% of all heinous crimes.
Problems / Predicted Reasons	• Cybercrime surge → lack of strong monitoring & awareness in online safety. • Robbery prevalence → public space safety & weak deterrence. • POCSO rise → inadequate child protection & awareness. • Murder/Burglary → localized law & order challenges.
Recommendations	• Cyber: Expand digital forensics, cyber law enforcement, and citizen training. • Robbery: Increase patrols, surveillance (CCTV), and hotspot policing. • POCSO: Strengthen child protection cells, school safety programs, and fast-track courts. • Burglary/Theft: Improve neighbourhood watch & night policing.



Top 10 Non-Heinous Crimes

Aspect	Details
Visual Type	Pie Chart

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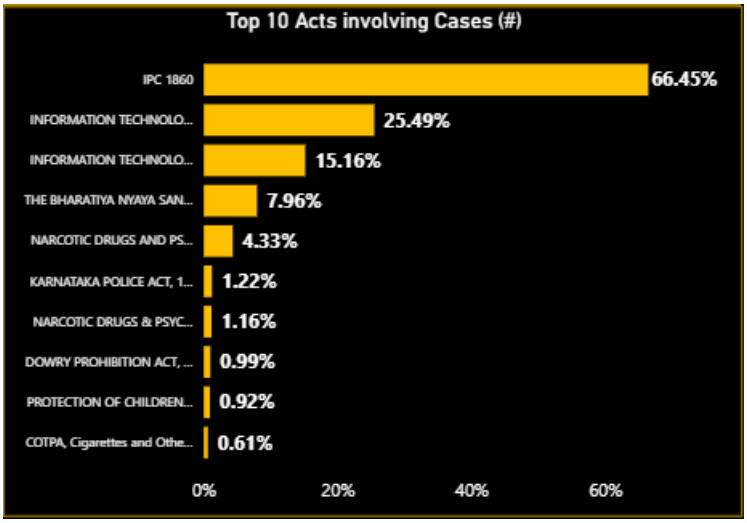
Data Fields	**Legend:** Crime Group Name **Values:** [Assigned Cases] **Filters:** FIR Type = Non-Heinous Crime Group Name Top 10 by Assigned Cases
Measures Used	[Assigned Cases]
Purpose	To highlight the most common non-heinous crimes and compare their share in overall distribution, focusing on high-frequency but less severe cases.
Insights	• Cyber Crime: 47.1% → almost half of all non-heinous cases. • Theft: 17.8%. • Missing Person: 11.9%. • Narcotic Drugs & Psychotropic Substances: 6.5%. • Cases of Hurt: 6.3%. • Cheating: 4.2%. • Local Act violations: < 2%.
Problems / Predicted Reasons	• Cybercrime dominance → weak cyber hygiene & lack of citizen awareness. • Theft & Missing Persons → high frequency creates police workload pressure. • Drugs → signals youth vulnerability and trafficking networks.
Recommendations	• Expand cyber awareness campaigns & IT policing cells. • Strengthen community policing & surveillance to reduce theft/missing persons. • Collaborate with health & social services to fight drug abuse. • Introduce preventive measures to reduce repeat petty crimes and backlog.



Top 10 Acts Involving Cases (%)

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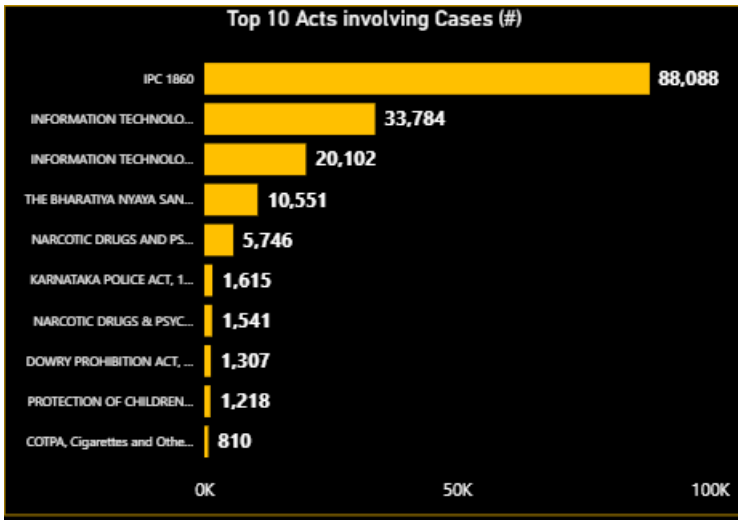
Aspect	Details
Visual Type	Clustered Bar Chart
Data Fields	**Y-axis (Categories):** Act_Description, Sec_Description **X-axis (Values):** [Assigned Cases Percentage] **Filter:** Act_Description Top 10 by [Assigned Cases Percentage]
Measures Used	[Assigned Cases Percentage]
Purpose	To show which legal Acts are most frequently invoked in FIRs. Provides insight into the legal areas where crime concentration is highest, guiding both law enforcement focus and policy reform priorities.
Insights	• IPC 1860 dominates → core of criminal justice system, applied to most FIRs. • IT Act 2000 → cyber fraud, hacking, identity theft; gaining importance in digital era. • IT Act 2008 (Amendment) → addresses cyber terrorism, phishing, data protection. • Bharatiya Nyaya Sanhita (BNS) 2023 → modern framework replacing parts of IPC, focusing on digital/organized crimes.
Problems / Predicted Reasons	• IPC's dominance overshadows smaller but critical Acts. • IT Act 2000 vs 2008 overlap may confuse reporting/analysis. • Only % share shown → actual case volumes not visible. • Lacks clarity on crime severity within Acts.
Recommendations	• Add dual-axis (count + %) for transparency. • Group IT Act 2000 & IT Act 2008 under a unified "Cybercrime Acts" category. • Build trend analysis by year for each Act → shows legal relevance evolution. • Apply color coding (IPC, IT, BNS) for readability. • Add drill-through by Sections to uncover case details (e.g., IPC 302 → murder cases).



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Top 10 Acts Involving Registered Cases

Aspect	Details
Visual Type	Clustered Bar Chart
Data Fields	**Y-axis:** Act Description, Section Description **X-axis:** Assigned Cases **Filters:** Top 10 Acts by Assigned Cases
Measures Used	[Assigned Cases]
Purpose	• Identify Acts of law that dominate registered cases. • Highlight the legal frameworks most invoked in FIRs. • Guide policy & judicial resource allocation where workloads are concentrated.
Insights	• IPC 1860: 88,088 cases → absolute majority. • IT Act 2000 (33,784) + IT Act 2008 (20,102) → cybercrime cluster rising fast. • Bharatiya Nyaya Sanhita (BNS) 2023: 10,551 cases, showing quick adoption post-reform. • NDPS Act (5,746) → persistent drug offences. • Other notable contributions: Karnataka Police Act (1,615), Dowry Prohibition Act (1,307), POCSO (1,218). • Together, IPC + IT Acts ≈ 70% of Top 10 Acts, showing concentration around general criminal and cyber domains.
Problems / Predicted Reasons	• Heavy IPC usage → backlog pressure on courts. • IT Act surge → cyber offences expanding beyond policing capacity. • NDPS/Dowry → entrenched social issues, not just legal gaps. • POCSO, though smaller in number, signals sensitive crimes needing faster disposal.
Recommendations	• Judicial fast-track for IPC-heavy caseloads. • Strengthen cyber forensic cells & IT law enforcement for IT Act cases. • Awareness + rehab programs for NDPS-related offences. • Expand child protection cells to monitor & speed up POCSO cases. • Legal modernization: simplify overlapping IPC sections, streamline BNS adoption.

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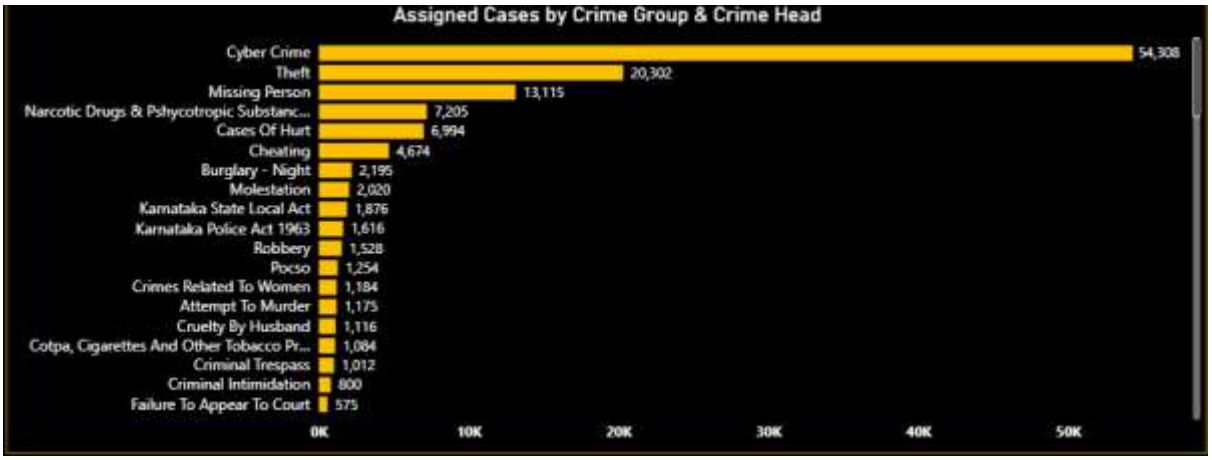

Assigned Cases by Crime Group & Crime Head

Aspect	Details
Visual Type	Clustered Bar Chart
Data Fields	**Y-axis:** CrimeGroup_Name, CrimeHead_Name **X-axis:** [Assigned Cases]
Measures Used	[Assigned Cases]
Purpose	• To analyze the distribution of assigned cases across different crime groups and heads. • To identify categories contributing the most to police workload.
Key Insights	• Cyber Crime: 54,308 cases (largest share). • Theft: 20,302 cases. • Missing Person: 13,115 cases. • NDPS-related crimes: 7,205 cases. • Cases of Hurt: 6,994 cases. • Failure to Appear in Court and Criminal Intimidation show relatively lower counts. • Overall, a few categories (Cyber, Theft, Missing Persons) dominate case load disproportionately.
Problems / Predicted Reasons	• Overcrowded chart: Too many categories → harder for users to interpret at once. • Small-value crimes invisible: Low-frequency bars overshadowed by major ones. • No % contribution: Raw counts only, no sense of proportion. - No time context: Users cannot see trends/seasonality.
Recommendations	• Add a Top 10 crimes filter to improve focus. • Create a 100% stacked bar view for proportion comparison. • Enable hierarchical drill-down (Crime Group → Crime Head → Crime Section).

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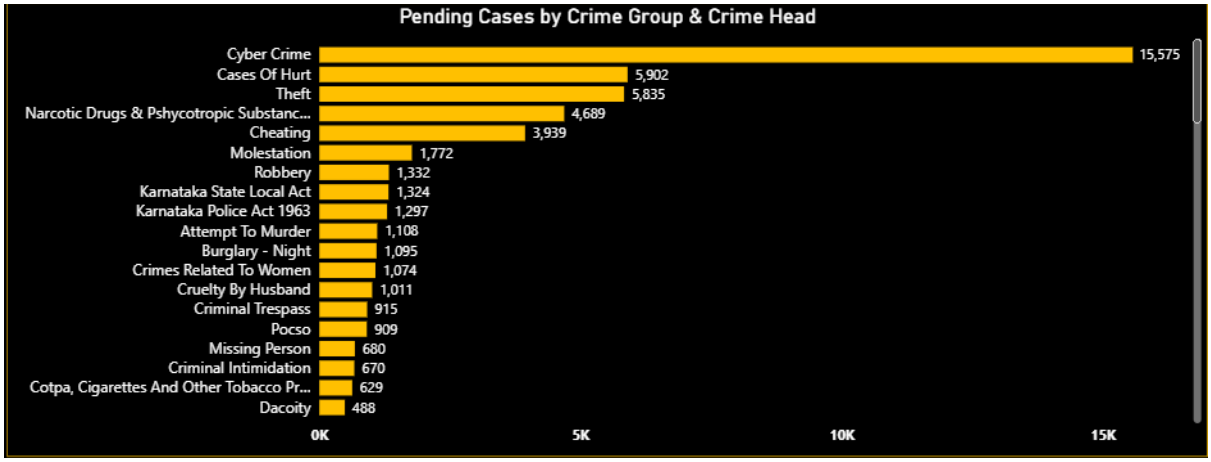
	• Add trend tooltips (monthly/yearly breakdown when hovering). • Include a benchmark line (e.g., average assigned cases per crime head) for better workload comparison.



Pending Cases by Crime Group & Crime Head

Aspect	Details
Visual Type	Clustered Bar Chart
Data Fields	**Y-axis (Categories):** CrimeGroup_Name, CrimeHead_Name **X-axis (Values):** [Pending Cases]
DAX Measures Used	[Pending Cases]
Purpose	• To provide an overview of backlog in cases, showing which crime groups/heads face delays. • To help prioritize resources for high-pending categories.
Key Insights	• Cyber Crime backlog is the largest among all categories. • Theft and Missing Person cases also show significant pending volumes. • NDPS Act and Cases of Hurt maintain steady pending loads, adding to the long-term backlog. • Lesser categories (Failure to Appear in Court, Criminal Intimidation) show smaller pending counts but may still create bottlenecks due to their sensitive nature. • Clear imbalance: a handful of categories dominate total pending cases.
Problems / Predicted Reasons	• Case overload in Cyber & Theft → Backlog worsens due to investigation complexity and evidence collection delays. • Court dependency → Cases stuck under "Pending@Court" inflate backlog, especially in NDPS and POCSO categories. • Visualization issue → Too many bars, making small pending categories invisible. • No temporal split, so users cannot track whether pending stock is increasing or reducing over time.

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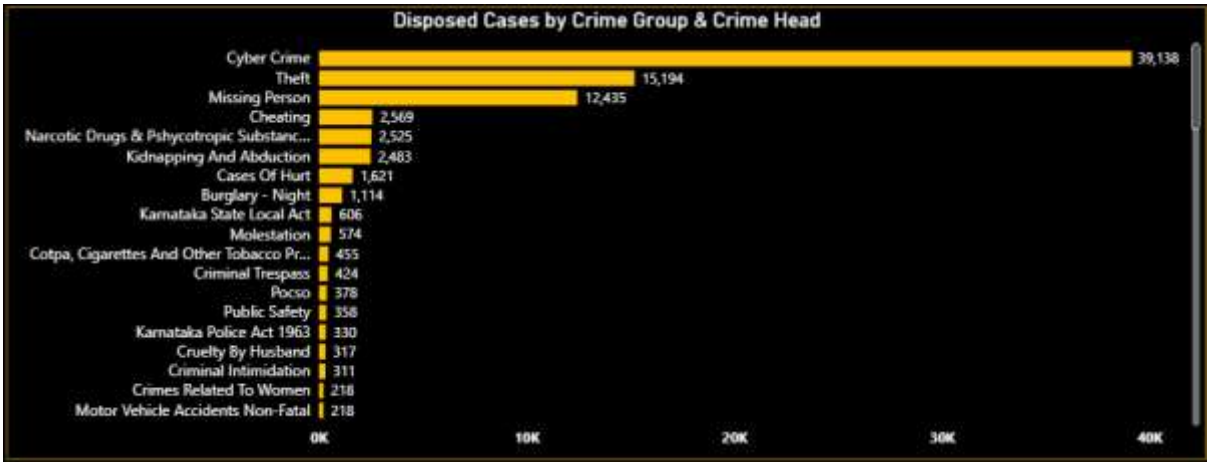
Recommendations	• Add a Top 10 Pending Crimes filter to declutter view. • Enable drill-through by Crime Group → Crime Head → Court/PS pending split. • Add trend over time to monitor backlog growth vs clearance. • Prioritize high-pending groups (Cyber, Theft, Missing Person) for targeted clearance drives.



Disposed Cases by Crime Group & Crime Head

Aspect	Details
Visual Type	Clustered Bar Chart
Data Fields	**Y-axis:** CrimeGroup_Name, CrimeHead_Name **X-axis:** [Cases Disposed]
Measures Used	[Cases disposed]
Purpose	• Track closure progress of cases across different crime groups. • Identify which crimes are resolved efficiently vs. lagging behind. • Provide performance insight on law enforcement workload handling.
Insights	• Cyber Crime: 39,138 disposed → highest, reflecting strong follow-up. • Theft: 15,194 disposals, second-highest. • Missing Person: 12,435 disposals → efficient clearance compared to pending load. • Low disposals in sensitive categories: Burglary at Night (1,114), Molestation (574), Motor Vehicle Accidents Non-Fatal (218). • Overall: focus remains on high-volume crimes, while smaller but critical categories lag.
Problems / Predicted Reasons	• Only absolute disposal counts shown → no efficiency % vs. assigned/pending. • No time-series → can't track disposal trend.

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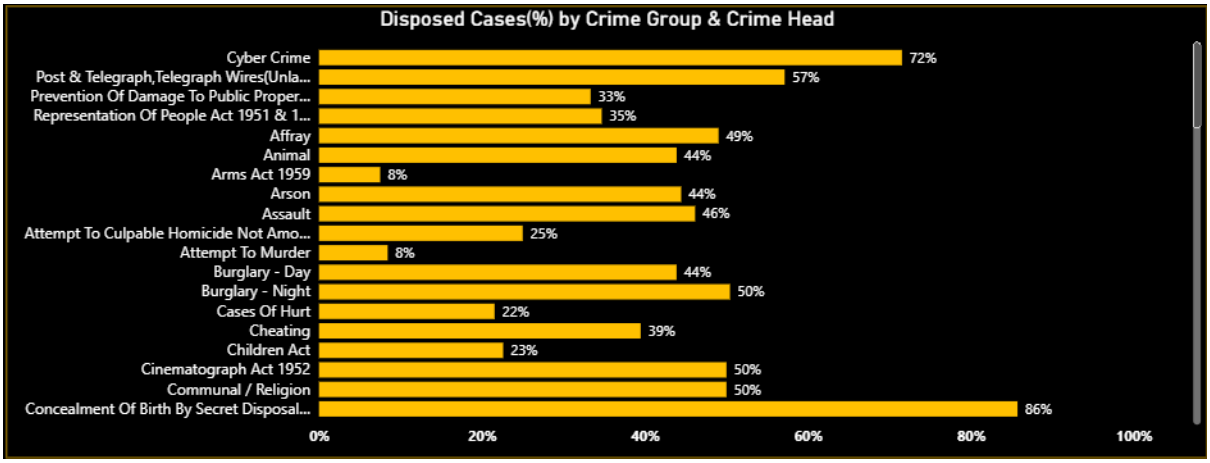
	• Sensitive cases (Molestation, Crimes Against Women) show very low disposals. • Lack of benchmark or ratio hides true clearance performance.



Disposed Cases (%) by Crime Group & Crime Head

Aspect	Details
Visual Type	Clustered Bar Chart
Data Fields	**Y-axis:** CrimeGroup_Name, CrimeHead_Name **X-axis:** [Disposed Rate]
Measures Used	[Disposed Rate]
Purpose	• Show what percentage of assigned cases are being disposed. • Compare clearance efficiency across crime categories.
Insights	• Concealment of Birth → 86% (highest). • Cyber Crime → 72% despite very high assigned cases (strong performance). • Post & Telegraph/Telecommunication → 57% (above average). • Children Act, Cinematograph Act, Religion crimes → ~50% (moderate). • Serious crimes like Attempt to Murder (25%), Culpable Homicide Attempt (25%), Arms Act (8%), Burglary (44%) show very poor clearance. • Crimes Against Women (e.g., Molestation) not visible → need deeper check.

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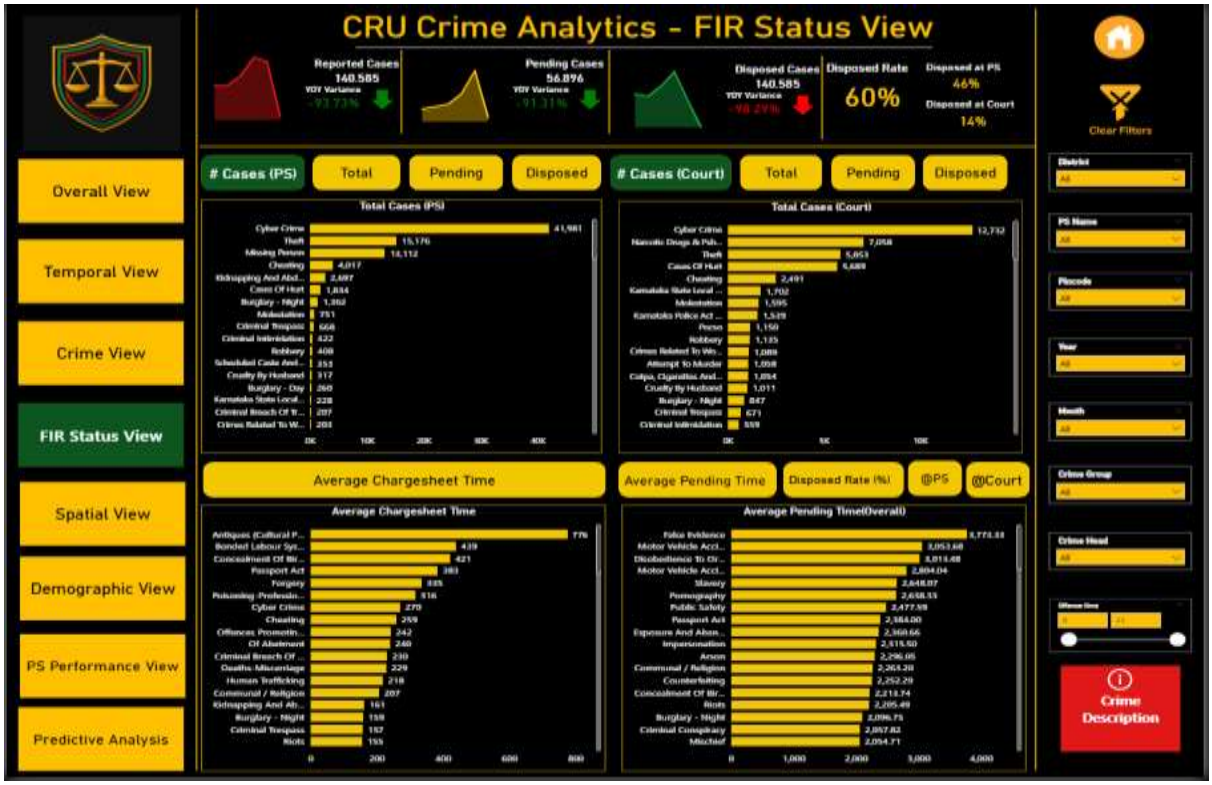

Button	Bookmark Triggered	Visuals Shown	Visuals Hidden	Notes
Victim Top 10	VictimTop10_BM	Top 10 victim categories (bar chart)	Victim Top 20, Victim Profile, Offender visuals	High-level victim view
Victim Top 20	VictimTop20_BM	Top 20 victim categories (bar chart)	Victim Top 10, Victim Profile, Offender visuals	More detail
Victim Profile	VictimProfile_BM	Victim demographic profile (Age, Gender, Occupation, etc.)	Victim Top 10, Victim Top 20	Structural demographics
Offender Top 10	OffenderTop10_BM	Top 10 offender categories	Offender Top 20, Offender Total, Victim visuals	High-level offender view
Offender Top 20	OffenderTop20_BM	Top 20 offender categories	Offender Top 10, Offender Total, Victim visuals	Detailed analysis
Offender Total	OffenderTotal_BM	Total accused count	Offender Top 10, Offender Top 20	Overall summary

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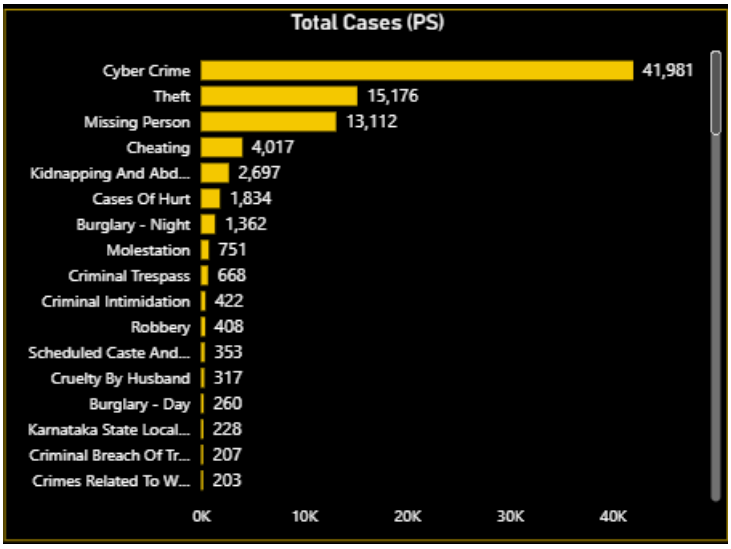

Total Cases Reported at Police Stations

Aspect	Details
Visual Type	Clustered Bar Chart
Data Fields	**Y-Axis:** CrimeGroup_Name, CrimeHead_Name **X-Axis:** [Total Cases (PS)]
Measures Used	[Total Cases (PS)]
Purpose	• Show total number of cases at Police Station (PS) level. • Identify which crime categories have the highest reporting volume.
Insights	• Cyber Crime → 41,981 cases (highest). • Theft (15,176), Missing Person (13,112), Cheating (4,017) are other major categories. • Missing Person cases form a large share but are mostly administrative, not criminal trials.
Problems / Predicted Reasons	• Cyber Crime and Theft show very high reporting, pointing to rising trends. • Missing Person cases inflate PS statistics without actual trial-based closure.
Recommendations	• Deploy specialized cyber cells to manage large Cyber Crime volumes. • Strengthen theft-prevention strategies. • Build tracking system for Missing Persons (closure reason, outcome monitoring).

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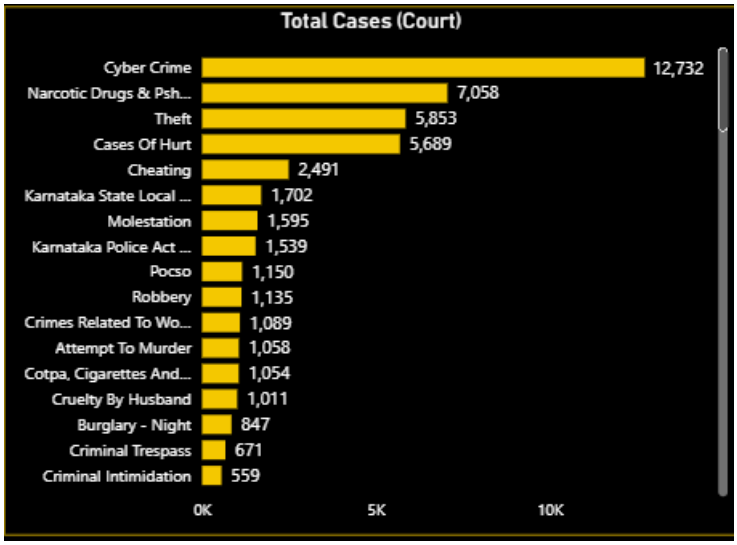

Total Cases Registered in Court

Aspect	Details
Visual Type	Clustered Bar Chart
Data Fields	**Y-Axis:** CrimeGroup_Name, CrimeHead_Name **X-Axis:** [Total Cases (Court)]
Measures Used	[Total Cases (Court)]
Purpose	• Show the total number of cases that reach the Court stage. • Compare which crime categories proceed from PS to trial.
Insights	• Cyber Crime → 12,732 cases (highest), but only ~30% of PS volume moves to Court. • Theft (7,058) and Narcotics (5,853) are other leading trial categories. • Missing Person cases are almost absent, as expected (administrative, not trial-bound).
Problems / Predicted Reasons	• Major drop from PS → Court in Cyber Crime and Cheating, suggesting weak follow-up or lack of evidence. • High Narcotics representation in Court may reflect focused enforcement, but could create backlog risk.
Recommendations	• Strengthen transfer tracking system (PS → Court). • Investigate why Cyber Crime and Cheating have low conversion rates (evidence gaps, digital expertise issues). • Monitor narcotics trials closely to manage backlog.

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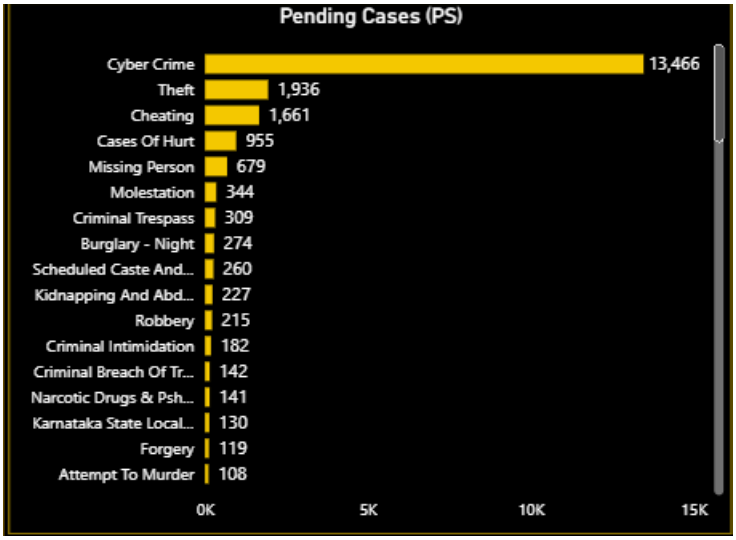

Pending Cases at Police Station (PS)

Aspect	Details
Visual Type	Clustered Bar Chart
Data Fields	**X-axis:** [Pending Cases (PS)] **Y-axis:** CrimeHead_Name
Measures Used	[Pending Cases (PS)]
Purpose	• Show volume of cases still pending at Police Station level. • Highlight categories with the highest backlog to guide resource allocation.
Insights	• Cyber Crime → 13,466 pending (highest). • Theft → 1,936; Cheating → 1,661; Hurt → 955. • Missing Person → 679 cases. • Other mid-range pendency: Molestation (344), Criminal Trespass (309), Burglary Night (274), SC/ST Atrocities (260), Kidnapping & Abduction (227), Robbery (215).
Problems / Predicted Reasons	• Cyber Crime backlog (13k+) signals severe investigation strain. • Theft, Cheating, and Hurt cases piling up → heavy PS-level workload. • Sensitive crimes (Missing Person, Kidnapping, Molestation) pending → direct public safety risk.
Recommendations	• Form Cyber Crime special task force for backlog clearance. • Prioritize pendency in Missing Person & Kidnapping to reduce victim risk. • Fast-track Theft and Cheating using digital evidence-based processes.

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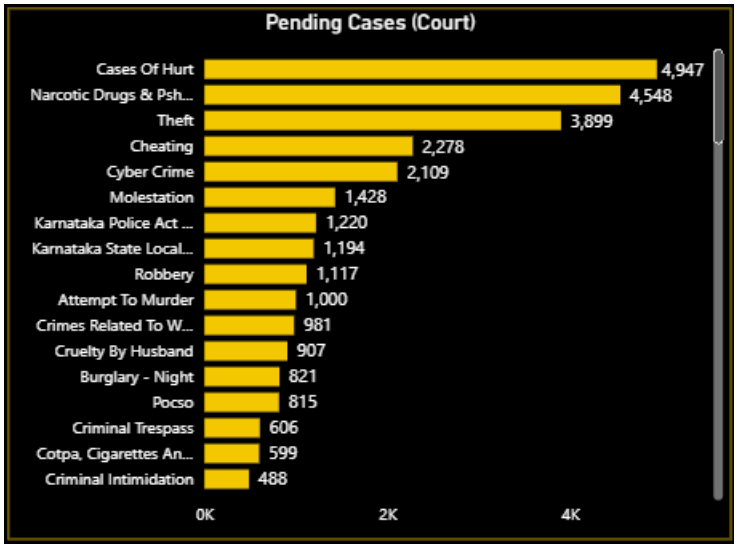

Pending Cases at Court

Aspect	Details
Visual Type	Clustered Bar Chart
Data Fields	**X-axis:** [Pending Cases (Court)] **Y-axis:** CrimeHead_Name
Measures Used	[Pending Cases (Court)]
Purpose	• Show categories with highest case pendency at the Court level. • Highlight backlog pressure points and sensitive case delays.
Insights	• Hurt → 4,947 (highest pendency). • NDPS → 4,548 (second-highest). • Theft (3,899), Cheating (2,278), Cyber Crime (2,109) form the next cluster of high-volume cases. • Molestation → 1,428. • Other significant pendency: Robbery (1,117), Attempt to Murder (1,100), Crimes Against Women (981), Cruelty by Husband (907), Burglary Night (821), POCSO (815).
Problems / Predicted Reasons	• Hurt & NDPS (~10k combined) dominate backlog → trials progressing very slowly. • Theft, Cheating & Cyber Crime → major economic crimes choking court time. • Sensitive cases (Molestation, POCSO, Women-related) pending → delay impacts justice and public trust.
Recommendations	• Set up NDPS-special courts to reduce narcotics backlog. • Use Lok Adalats/plea bargaining for petty theft to reduce trial load.

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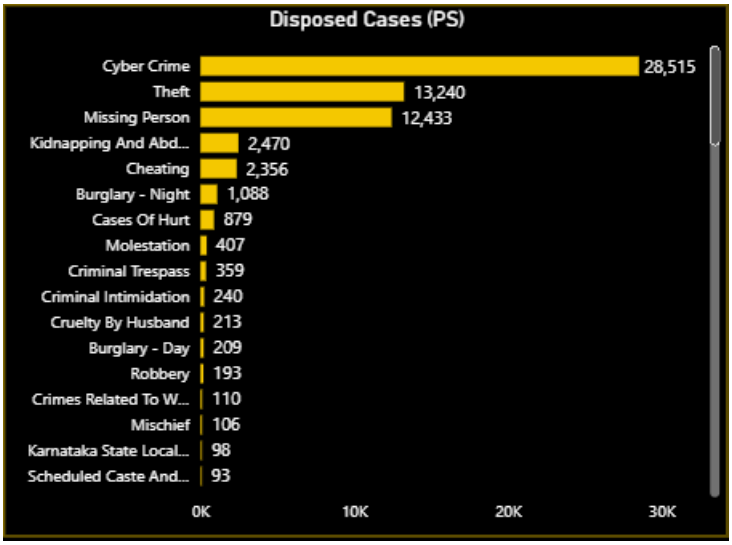
	• Introduce fast-track courts for Women & POCSO crimes. • Promote digital evidence-based fast disposal for Cyber & Cheating cases.



Disposed Cases at PS

Aspect	Details
Visual Type	Clustered Bar Chart
Data Fields	**X-axis:** [Disposed Cases (PS)] **Y-axis:** CrimeGroup_Name, CrimeHead_Name
Measures Used	[Disposed Cases (PS)]
Purpose	• Analyze which crimes are being disposed at Police Station (PS) level. • Identify which crime categories close quickly at PS vs. which linger. • Assess PS efficiency in handling case closures.
Insights	• Cyber Crime → 28,515 disposals (highest). • Theft (13,240) and Missing Person (12,433) also high, showing strong PS-led resolution. • Medium disposals: Kidnapping & Abduction (2,470), Cheating (2,356). • Low disposals: Molestation (407), Cruelty by Husband (213), Robbery (193). - SC/ST cases → only 93 disposed, usually escalated.
Problems / Predicted Reasons	• High disposal but weak court conversion (e.g., Cyber Crime, Theft) may indicate premature PS closure. • Sensitive crimes (Molestation, Domestic Violence, Robbery) show very low disposals. • SC/ST-related cases often escalated, negligible PS disposal.

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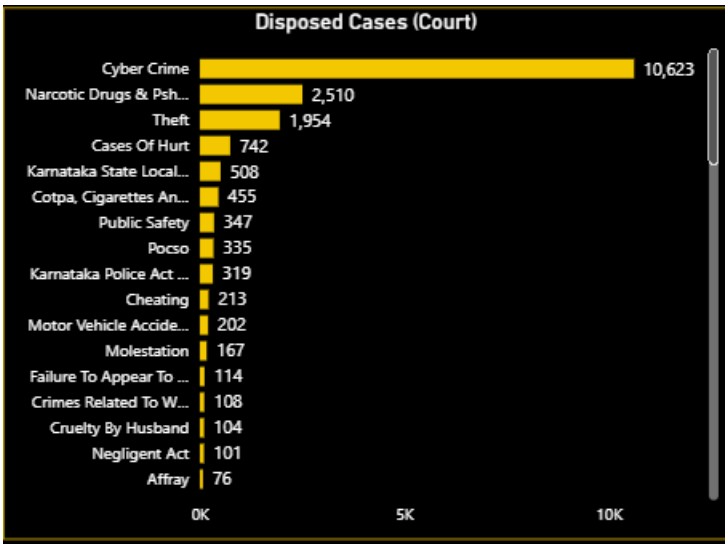
Recommendations	• Audit high-volume PS disposals (Cyber Crime, Theft, Missing Person) to ensure genuine resolution. • Create special handling units for sensitive crimes (Molestation, Domestic Violence, SC/ST). • Train PS staff for better legal handling so disposals withstand court scrutiny. • Policy review to prevent PS-level closures bypassing due judicial process.



Disposed Cases at Court

Aspect	Details
Visual Type	Clustered Bar Chart
Data Fields	**X-axis:** [Disposed Cases (Court)] **Y-axis:** CrimeHead Name
Measures Used	[Disposed Cases (Court)]
Purpose	• Show the volume of cases disposed at the Court level. • Identify which crime categories are seeing strong judicial resolution vs. weak disposal.
Insights	• Cyber Crime → 10,623 disposals (highest, strong focus). • NDPS → 2,510; Theft → 1,954; Hurt → 742. • Medium disposals: Karnataka State Local Acts (508), COTPA (455), Public Safety (347), POCSO (335). • Very low disposals: Molestation (167), Cheating (213), Crimes Against Women (108), Cruelty by Husband (104). • Other minor disposals: Negligent Act (101), Motor Vehicle Accident (202), Affray (76).
Problems / Predicted Reasons	• High Cyber Crime disposals but pendency (2,109) remains → heavy inflow of cases. • NDPS, Theft, Hurt → large disposals but even bigger backlog (e.g., NDPS: 4,548 pending vs. 2,510 disposed).

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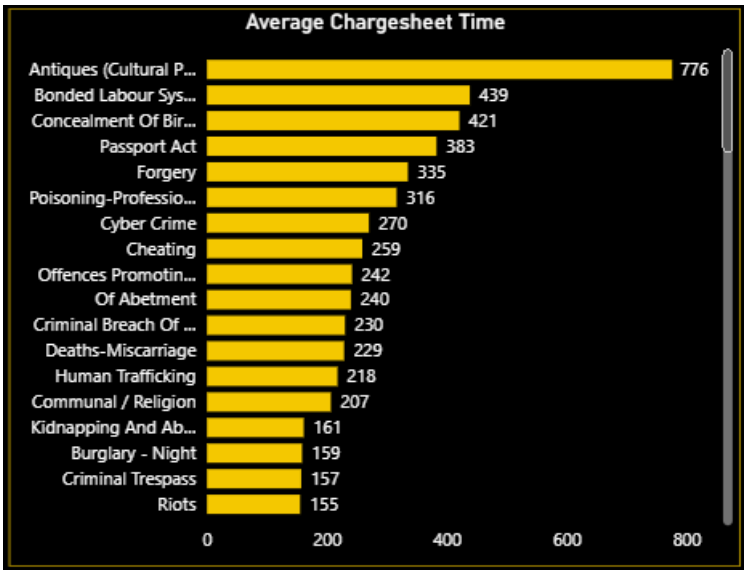
	• Sensitive crimes (POCSO, Molestation, Women-related) show very low disposals, raising justice delay concerns.
Recommendations	• Cyber Crime: Continue fast-track disposal, but also invest in preventive measures (digital policing, awareness, tech-enabled investigation). • NDPS: Expand dedicated special courts; backlog exceeds progress. • Women & Child crimes (POCSO, Molestation, Cruelty): Mandate time-bound hearings to protect victims. • Theft & Cheating: Adopt summary trials or Lok Adalats for quicker disposal.



Average Chargesheet Time

Aspect	Details
Visual Type	Clustered Bar Chart
Data Fields	**X-axis:** Average Chargesheet Time **Y-axis:** CrimeGroup_Name, CrimeHead_Name
Measure Used	[Average Chargesheet Time]
Purpose	Show avg. time (days) to file chargesheets across crimes; identify slow vs. fast categories.
Insights	• Very high: Cultural Property (776d), Bonded Labour (439d), Concealment of Birth (421d). • Moderate: Forgery (335d), Cyber Crime (270d), Cheating (259d). • Low: Burglary (155d), Trespass (157d), Kidnapping (161d).
Problems	• Long delays in cultural, labour & birth concealment → low priority/complex cases. • Cyber & fraud cases delayed → weak evidence handling.
Recommendations	Fast-track units for high-delay crimes, digitization of evidence, prioritization framework, better cross-agency coordination.

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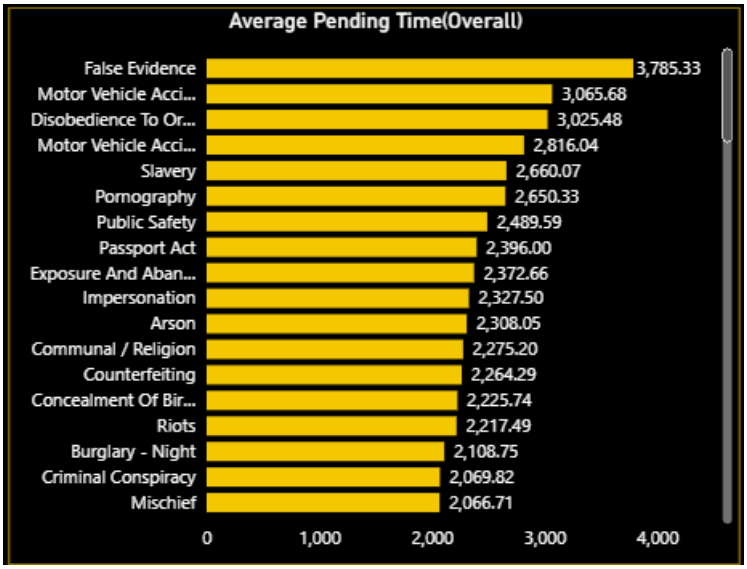

Average Pending Time (Overall)

Aspect	Details
Visual Type	Clustered Bar Chart
Data Fields	**X-axis:** [Average Pending Time] **Y-axis:** CrimeGroup_Name, CrimeHead_Name
Measure Used	[Average Pending Time]
Purpose	Show avg. days cases remain pending in courts; highlight judicial delays and efficiency gaps.
Insights	• Very high: Arson (2,286d), Burglary-Night (2,086d), Public Property Damage (2,030d), Assault (1,876d), Animal Cruelty (1,888d). • Cyber Crime ~1,754d (≈5 yrs). • Moderate: Attempt to Murder (1,702d), Adulteration (1,171d), Bonded Labour (1,162d). • Lower: Antiques (677d), Post & Telegraph (728d).
Problems	• Cyber Crime → 5 yrs pendency, risk of evidence loss. • Violent crimes (Assault, Attempt to Murder, Arson) delayed → safety risk. • Governance-related crimes (Adulteration, Bonded Labour) dragged for 3–4 yrs.

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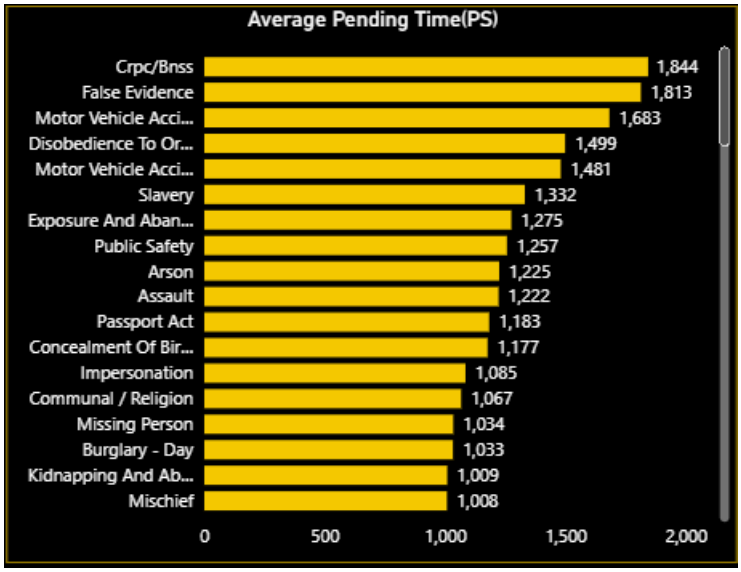
Recommendations	Fast-track courts for cyber/violent/economic crimes, AI-based case scheduling, digital evidence systems, prioritize violent crime trials, ADR methods for minor cases.



Average Pending Time (PS)

Aspect	Details
Visual Type	Clustered Bar Chart
Data Fields	**X-axis:** [Average Pending Time (PS)] **Y-axis:** CrimeGroup_Name, CrimeHead_Name
Measure Used	[Average Pending Time (PS)]
Purpose	Track avg. days cases remain pending at Police Stations; highlight slow investigations & resource gaps.
Insights	• Very high: CrPC/BNSS (1833d), False Evidence (1802d), MVAs (1488–1470d). • High: Slavery (1321d), Assault (1211d), Arson (1214d), Passport Act (1172d). • Moderate: Communal/Religion (1056d), Missing Person (1023d), Burglary-Day (1022d). • Lower: Kidnapping (998d), Mischief (997d).
Problems	• CrPC/BNSS & False Evidence → 5 yrs backlog. • MVAs delayed 4 yrs → affects victims' compensation. • Serious crimes (Slavery, Assault, Arson) pending 3+ yrs.

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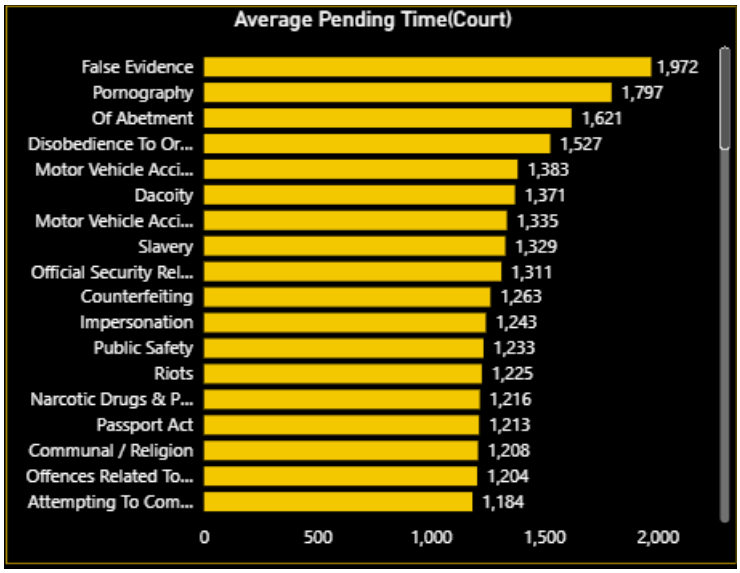
	• Sensitive cases (Missing, Kidnapping, Communal) also >1000d.
Recommendations	Special cells for MVAs & evidence-heavy cases, streamline forensic/medical processes, digitize accident claims, monitor sensitive cases closely, regular PS-level performance reviews.



Average Pending Time (Court)

Aspect	Details
Visual Type	Clustered Bar Chart
Data Fields	**X-axis:** Average Pending Time (Days) **Y-axis:** CrimeHead_Name
Measure Used	Average Pending Time (Court)
Purpose	Track avg. pending time in courts; highlight systemic judicial delays; assess efficiency of case disposal.
Insights	• Very high pendency: False Evidence (1962d), Pornography (1787d), Abetment (1611d), Disobedience to Orders (1517d). • Long delays in MVAs (1325–1373d), Dacoity (1361d), Slavery (1319d). • Sensitive crimes delayed: Riots (1215d), NDPS (1206d), Women-related offences (1194d), Communal crimes (1198d).
Problems	• Sensitive/victim-related crimes (Pornography, Women, Riots, Communal, MVAs) delayed 3–5 yrs.

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	• Security & narcotics cases also backlogged. • False Evidence worst hit (5+ yrs).
Recommendations	• Fast-track courts for Pornography, Women, Riots, Communal crimes. • Shift accident claims to tribunals. • Special NDPS benches; digital evidence mgmt. • ADR (Lok Adalats) for minor/financial offences. • AI-based scheduling & digitization for backlog reduction.

Average Pending Time (Court)

Aspect	Details
Visual Type	Clustered Bar Chart
Data Fields Used	**X-axis:** Average Pending Time (Days) **Y-axis:** CrimeHead_Name
DAX Measures Used	[Average Pending Time (Court)]
Purpose	• Shows average time cases remain pending in court. • Highlights systemic delays. • Tracks efficiency of case disposal.
Key Insights	• Longest delays: False Evidence (1962d), Pornography (1787d), Abetment (1611d). • Victim-related: Motor Accidents (1325–1373d), Women-related offences (1194d). • Security-related: Riots (1215d), Communal crimes (1198d), NDPS (1206d). • Other long delays: Slavery, Counterfeiting, Impersonation (>3 yrs).
Problem Areas	• Sensitive crimes (Pornography, Women, Riots, Communal) pending 3–5 yrs. • Victim-related cases (Accidents, Women) cause social & financial harm. • Drug & security offences delayed → risk to society.
Recommendations	• Fast-track courts: Pornography, Women, Riots, Communal. • Tribunals: Motor accident claims. • Specialised benches: NDPS courts; digital evidence for cyber & pornography. • ADR: Counterfeiting, impersonation → Lok Adalats. • Technology: AI-based scheduling; digitize backlog-heavy cases.

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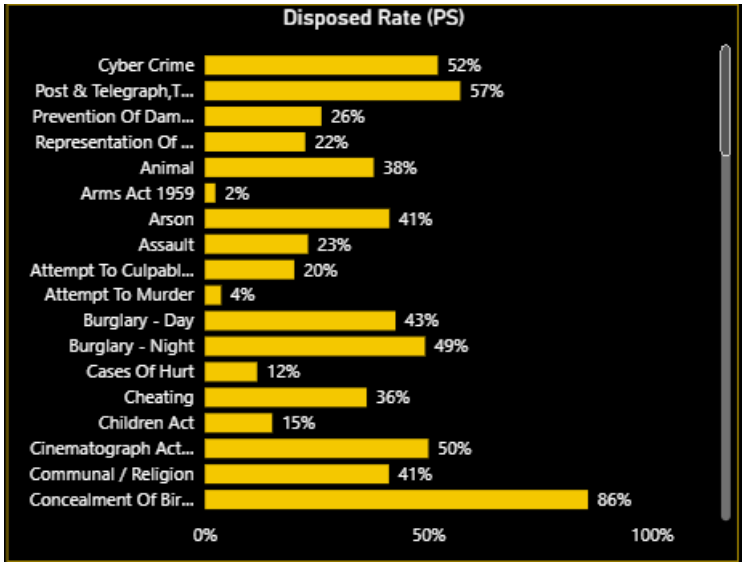

Disposed Rate (PS)

Aspect	Details
Visual Type	Clustered Bar Chart
Data Fields Used	**X-axis:** [Disposed Rate PS] **Y-axis:** CrimeHead_Name
DAX Measures Used	[Disposed Rate (PS)]
Purpose	• To measure disposal efficiency of cases at the Police Station (PS) level. • Helps identify strong vs weak enforcement areas. • Highlights crime categories with high vs low clearance.
Key Insights	• Highest disposal: Concealment of Birth (86%), Post & Telegraph Act (57%), Cybercrime (52%). • Moderate: Burglary Day (43%), Communal Offences (41%), Arson (41%). • Low: Cheating (36%), Assault (23%), Hurt (12%), Children Act (15%). • Critical concern: Attempt to Murder (4%), Arms Act (2%) – almost stalled.
Problem Areas	• Wide variance → 2% (Arms Act) vs 86% (Concealment). • Heinous offences (Attempt to Murder, Arms Act, Children Act) poorly disposed. • Common crimes (Assault, Hurt, Cheating) under-disposed (12–36%). • Regulatory/technical cases (Cybercrime, Post & Telegraph) cleared better.
Recommendations	• Prioritize heinous crimes → fast-track Attempt to Murder, Arms Act, Children Act. • Strengthen evidence → medico-legal, digital & witness handling training.

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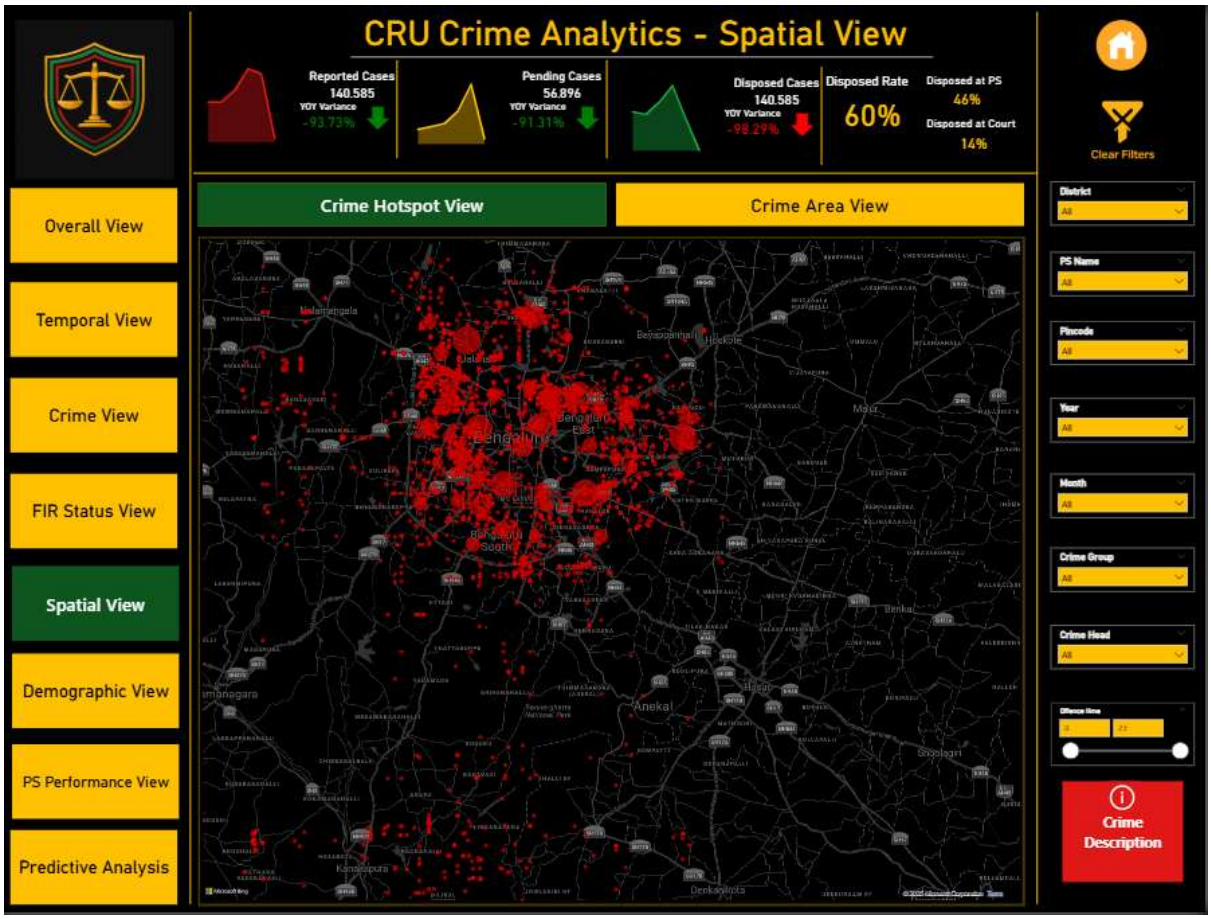
	• Replicate success → cybercrime cells model for violent crimes. • Accountability → weekly monitoring at DCP/SP level for low-disposal categories.

CRU Crime Analytics – Spatial View



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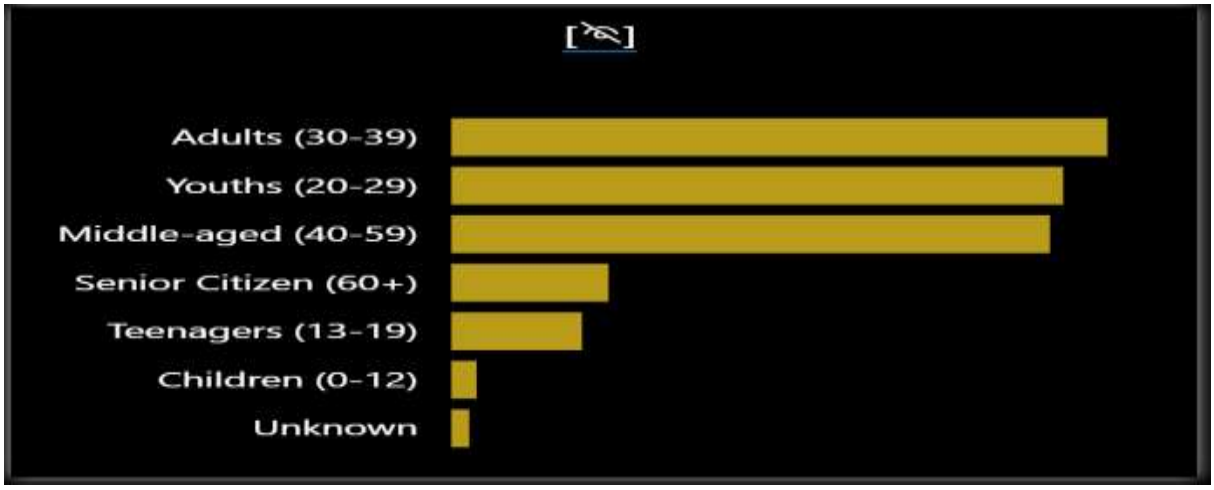
Gender wise reported cases Tooltip

Aspect	Details
Visual Type	Donut Chart (Tooltip Page)
Data Fields Used	Legend: Gender Values: Reported Cases

DAX Measures Used	Active UNIT =
	`VAR _unit = SELECTEDVALUE(Fact_Crime[UnitName])`
	`VAR _Gender = SELECTEDVALUE('victim Sex Order'[Gender])`
	`RETURN`
	`IF(_unit=BLANK(),_gender,_unit)`
	Used to dynamically display Unit Name in tooltip.

Purpose	• Provide demographic insight into reported cases by gender. • Enhance map analysis with additional context when hovering over a bubble. • Help compare male vs female victim distribution per Police Station (PS).

Tooltip Page – Age Segment–wise Reported Cases



Aspect	Details
Visual Type	Clustered Bar Chart (Tooltip Page)
Data Fields Used	X-axis: Reported Cases Y-axis: Age Segment (e.g., Child, Youth, Adult, Senior)

DAX Measures Used	Active UNIT =
	`VAR _unit = SELECTEDVALUE(Fact_Crime[UnitName])`
	`VAR _Gender = SELECTEDVALUE('victim Sex Order'[Gender])`
	`RETURN`
	`IF(_unit=BLANK(),_gender,_unit)`
	
	Used to dynamically display Unit Name in tooltip.

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Purpose	• Provide demographic breakdown of reported cases by age group. • Support hotspot and workload analysis with victim profile insights. • Compare case distribution across age segments for each Police Station.

Crime Hotspot View

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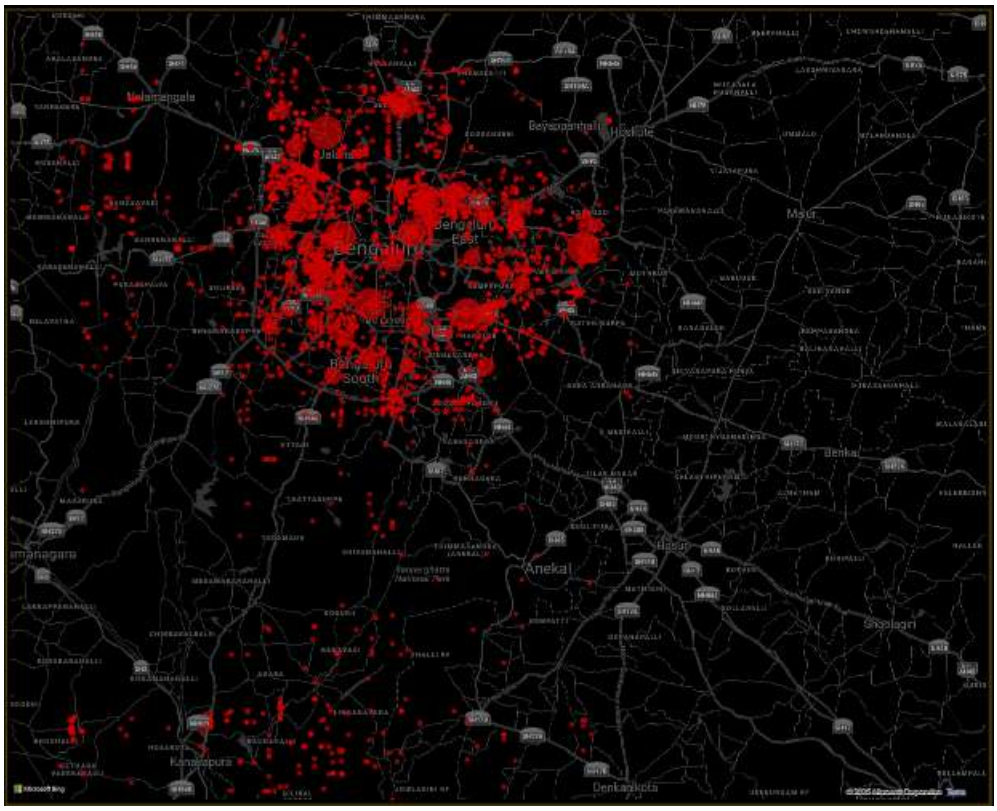

Aspect	Details
Visual Type	Map (Bubble Layer – Cluster Plot)
Data Fields Used	Latitude, Longitude, Bubble Size → Assigned Cases, Tooltip → Unit Name
DAX Measures Used	[Assigned Cases] (calculated measure for case allocation per PS/Unit)
Tooltip	• Unit Name – Police Station handling the cases. • Assigned Cases – total workload of that station. • Gender-wise Reported Cases (from tooltip page) – breakdown of male vs female victims. • Dynamic Filters Applied – Pincode, Crime Group, Crime Head, Year/Month.
Purpose	• To visualize spatial distribution of assigned cases. • Identify crime hotspots and regional concentration. • Compare workload intensity across Police Stations (PS).
Key Insights	• Major clustering visible in central Bengaluru – indicates high crime density. • Peripheral regions (Anekal, Ramanagara, Kanakapura) show smaller but spread-out cases. • Certain PS units manage disproportionately higher caseloads. • Tooltip reveals *gender bias* in reported cases, aiding demographic insights.
Problem Areas	• Overburdened PS units in central city – may strain enforcement capacity. • Rural/outer PS units show fewer but scattered cases – may cause

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	resource allocation inefficiency. • Gender-wise data shows imbalance (e.g., higher reporting among women in urban PS, but underreporting in rural areas).
Recommendations	• Deploy additional staff/resources to PS with highest bubble density. • Use hotspot data to plan targeted patrols in high-crime pockets. • Address gender-based crime trends → women helpdesks, awareness in rural areas. • Leverage peripheral PS with spare capacity to support overburdened ones.

Aspect	Details
Visual Type	Tree Map
Data Fields Used	Category: District_Name, UnitName, Village Values: Assigned Cases
Purpose	• To compare the proportion of total cases across different PS/Units. • Quickly highlight which PS contributes most to the total caseload. • Provide a hierarchical view (District → Unit → Village).
Key Insights	• Bengaluru City occupies the largest block, indicating significantly higher caseload than other districts. • Other districts/PS appear smaller, showing lesser contribution to overall cases.

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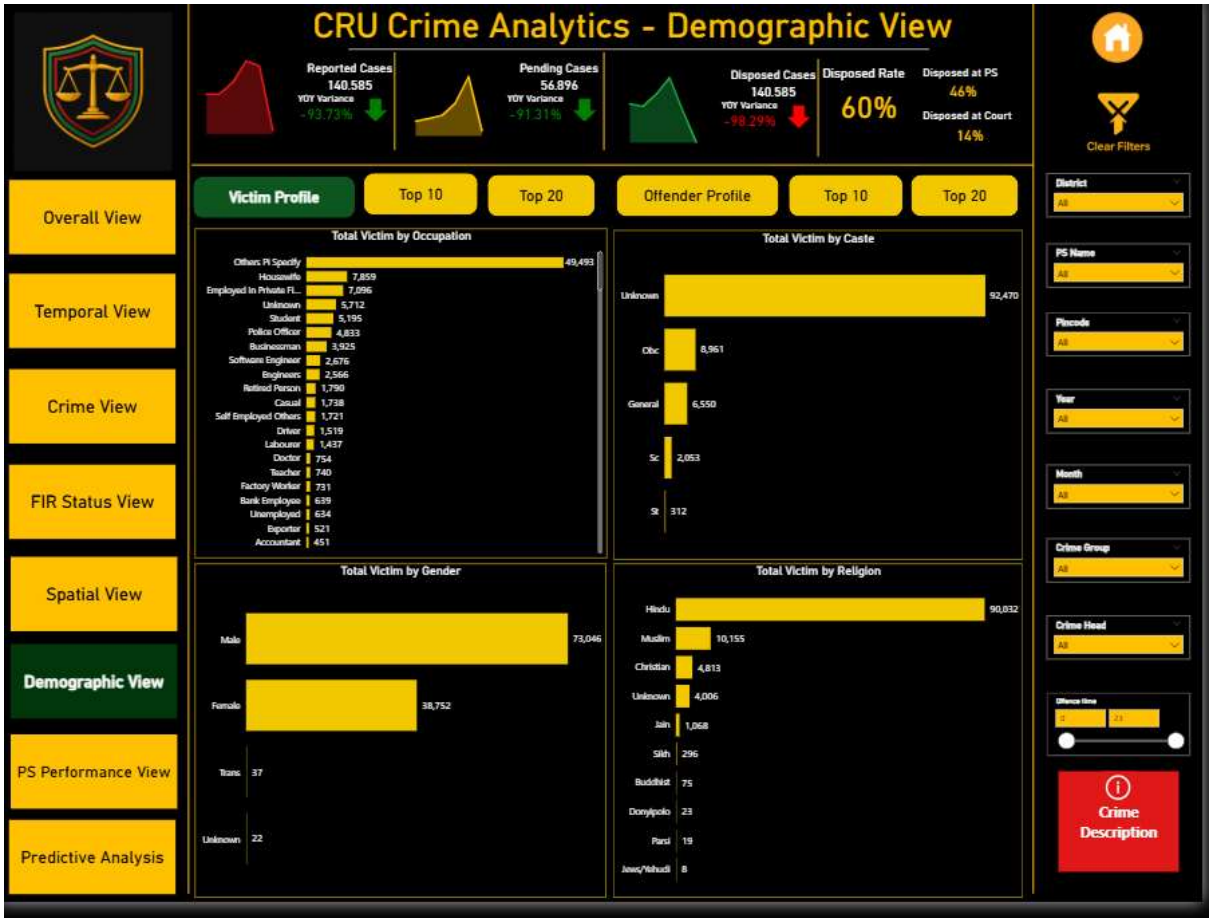
	• Visual imbalance indicates crime is not evenly distributed across PS units.
Problem Areas	• Central urban PS are heavily loaded, creating enforcement strain. • Rural/outer PS show low but scattered case distribution. • Lack of case distribution balance between urban and rural units.
Recommendations	• Prioritize staffing in Bengaluru City PS to manage high workload. • Investigate underlying reasons (population density, commercial areas, hotspots) behind the high case volume. • Redistribute cases/resources to underutilized PS where possible. • Use the treemap to regularly monitor changes in caseload distribution.



Victim Profile

Aspect	Details
Visual Type	Multiple Clustered Bar Charts (Occupation, Caste, Gender, Religion)
Data Fields Used	**Y-axis :** [Victim Occupation],[Victim Caste],[Victim Gender],[Victim Religion] **X-axis :** [Assign Cases]

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DAX Measures Used	[Assign Cases]
Purpose	• To analyze crime victims' profiles across socio-demographic attributes. • Identifies vulnerable groups by occupation, caste, gender, and religion. • Supports resource planning, victim support services, and targeted interventions.
Key Insights	• By Occupation: Highest → Others (PS Ready) (49,493 victims), followed by Housewives (7,859) and Private Employees (7,056). • By Caste: Majority recorded as Unknown (92,470), followed by OBC (8,961), General (6,550). • By Gender: Males (73,046) nearly double females (38,752); Transgender (37) cases minimal. • By Religion: Hindus (90,032) most affected, followed by Muslims (10,155), Christians (4,813).
Problem Areas	• Extremely high Unknown caste (92k+) → data quality gap. • Gender skew → majority male victims, but female vulnerability (esp. housewives) also significant. • Occupational category Others (PS Ready) too broad → needs standardization. • Religion-wise → over-representation of Hindu victims (needs normalization with population share).
Recommendations	• Improve data entry practices → reduce "Unknown" caste/religion fields. • Disaggregate "Others (PS Ready)" into meaningful categories. • Special victim support for vulnerable groups (housewives, private employees, students). • Gender-sensitive policing programs (esp. women safety & awareness). • Cross-tab analysis of religion × occupation and gender × occupation for deeper insights.

Offender Profile

Aspect	Details
Visual Type	Multiple Clustered Bar Charts (Occupation, Caste, Gender, Religion)
Data Fields Used	**Y-axis :** [Offender Occupation],[Offender Caste],[Offender Gender],[Offender Religion] **X-axis :** [Assign Cases]
DAX Measures Used	[Assign Cases]
Purpose	• To analyze crime offenders' socio-demographic distribution.

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	• Helps identify high-risk groups contributing to crimes. • Supports targeted preventive measures, awareness drives, and rehabilitation programs.
Key Insights	• By Occupation: Majority Unknown (75,591), followed by Others (28,809) and Businessmen (4,173). • By Caste: Large portion Unknown (109,771), followed by OBC (4,944), SC (2,434), General (1,985). • By Gender: Males (67,775) dominate → more than 5x females (11,823). • By Religion: High Unknown (75,007), Hindu offenders (39,705), Muslims (12,528), Christians (3,567).
Problem Areas	• Excessive "Unknown" entries in caste & occupation → limits analysis quality. • Gender imbalance → indicates crime is male-dominated. • Occupational grouping "Others" too broad → lacks actionable insights. • Religion data missing for ~75k offenders (major gap).
Recommendations	• Strengthen data collection standards to reduce "Unknown" values. • Break down "Others" occupation category into sub-segments (students, unemployed, daily wage, etc.). • Implement male-focused crime prevention programs and awareness campaigns. • Cross-tab offender profile with victim demographics → identify patterns like same caste/religion conflicts or occupation-driven crimes.

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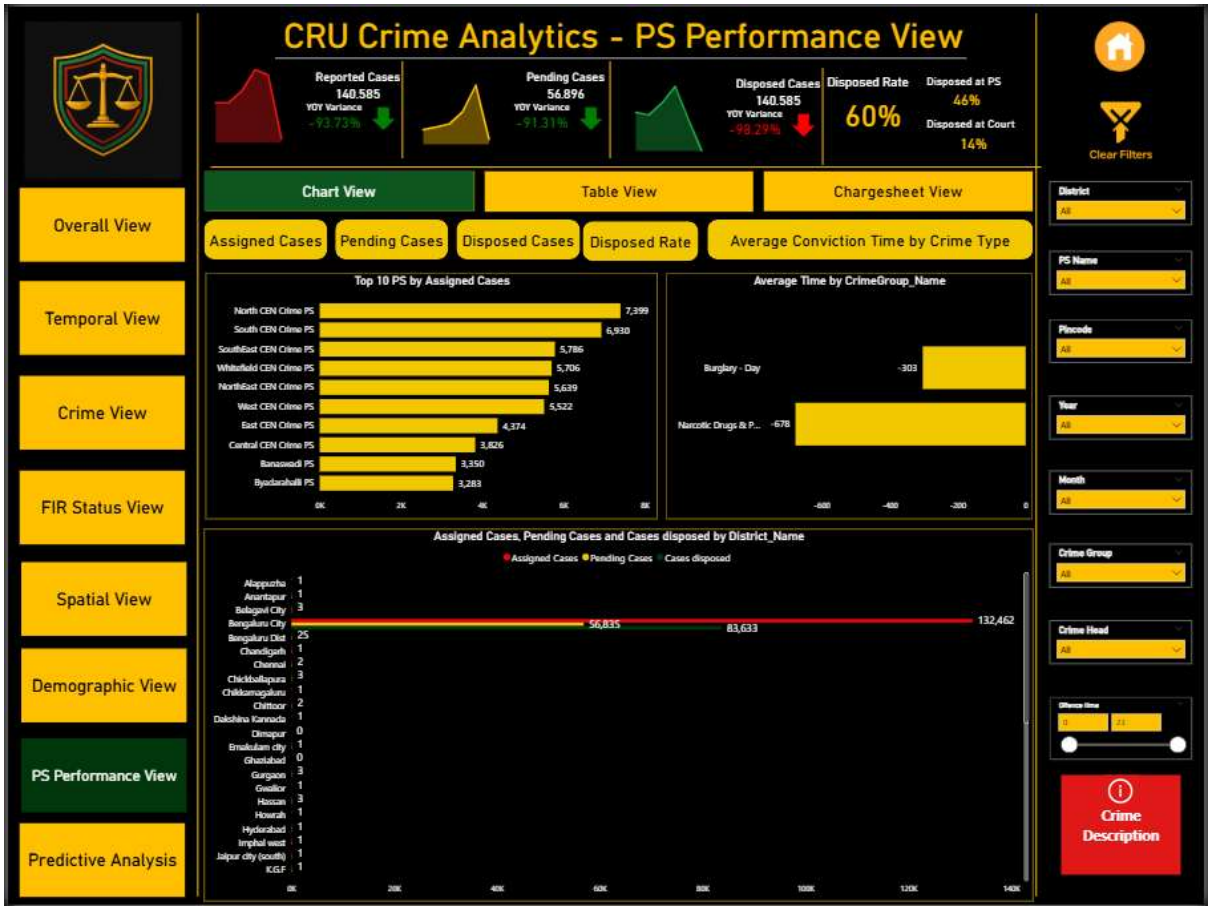

Performance View – Main Sections

Button	Bookmark Triggered	Visuals Shown	Visuals Hidden	Notes
Chart View	PSPerfChart_BM	PS performance charts (Pending, Disposal, etc.)	Table View, Chargesheet View	Default section
Table View	PSPerfTable_BM	Assigned vs Pending vs Disposed (tabular)	Chart View, Chargesheet View	Detailed table comparison
Chargesheet View	Chargesheet_BM	Avg Chargesheet Filing Time chart	Chart View, Table View	Focus on judicial delays

PS Performance – Chart View (Sub-Buttons)

Button	Bookmark Triggered	Visuals Shown	Visuals Hidden	Notes
Assign Cases	AssignCases_BM	Assigned cases per PS (bar chart)	Other Chart View visuals	Shows workload
Pending Cases	PendingCases_BM	Pending cases per PS	Others	Highlights backlog

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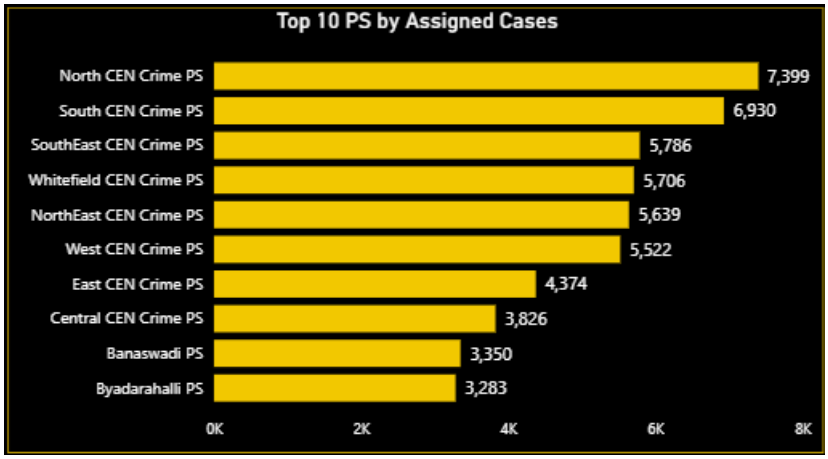
Disposed Cases	DisposedCases_BM	Disposed cases per PS	Others	Cleared cases
Disposal Rate	DisposalRate_BM	Disposal % per PS	Others	KPI vs benchmark (≥70%)
Average Conviction Time	AvgConviction_BM	Avg conviction time per PS	Others	Delay monitoring



Top 10 PS by Assigned Cases

Aspect	Details
Visual Type	Clustered Bar Chart
Data Fields	**X-axis:** [Assigned Cases] **Y-axis:** UnitName **Filter:** Unit Name Top 10 by Assign Cases
Measures Used	[Assigned Cases]
Purpose	To identify which Police Stations (PS) receive the highest number of cases, showing workload distribution.
Insights	• North CEN PS (7,399) and South CEN PS (6,930) have the highest assigned cases. • Other busy PS include Southeast (5,786), Whitefield (5,706), and Northeast CEN (5,629). • Smaller PS like Byadarahalli (3,283) have much lower case assignments.
Problems / Limitations	• High assigned cases don't show efficiency — a PS could have many cases but poor disposal. • Data does not account for manpower size, so workload fairness is unclear.
Recommendations	• Allocate more manpower/resources to high-load PS. • Monitor disposal % alongside assignments for true efficiency.

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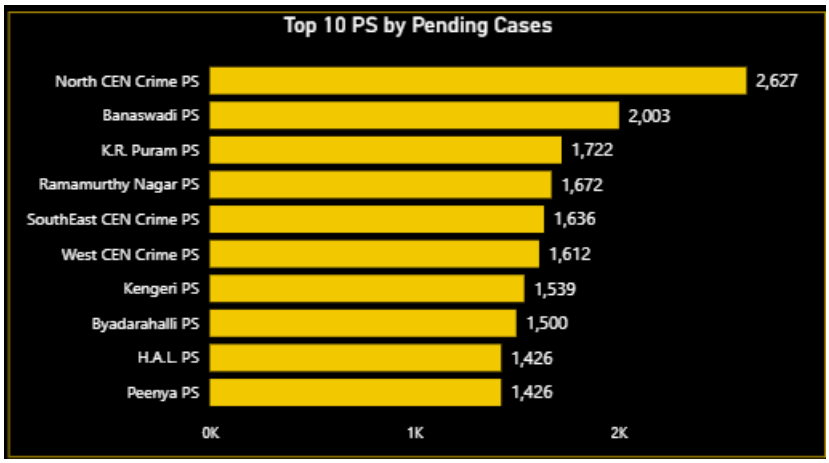
	• Review workload distribution across PS to ensure balance.



Top 10 PS by Pending Cases

Aspect	Details
Visual Type	Clustered Bar Chart
Data Fields	**X-axis:** [Pending Cases] **Y-axis:** UnitName **Filter:** UnitName Top10 by pending Cases
Measures Used	[Pending Cases]
Purpose	To highlight which Police Stations have the most backlog, indicating operational bottlenecks.
Insights	• North CEN PS (2,627) and Banashankari PS (2,003) lead in pendency. • West CEN (1,612), Byadarahalli (1,563), and Northeast CEN (1,300) also show significant pending loads. • Smaller PS pendency is lower (<1,200 cases).
Problems / Limitations	• High pendency indicates slow investigation or resource gaps. • Does not show pendency age (new vs. old cases).
Recommendations	• Prioritize pendency clearance in high-backlog PS. • Introduce fast-track review for >1000-day pending cases. • Match case load with available investigators.

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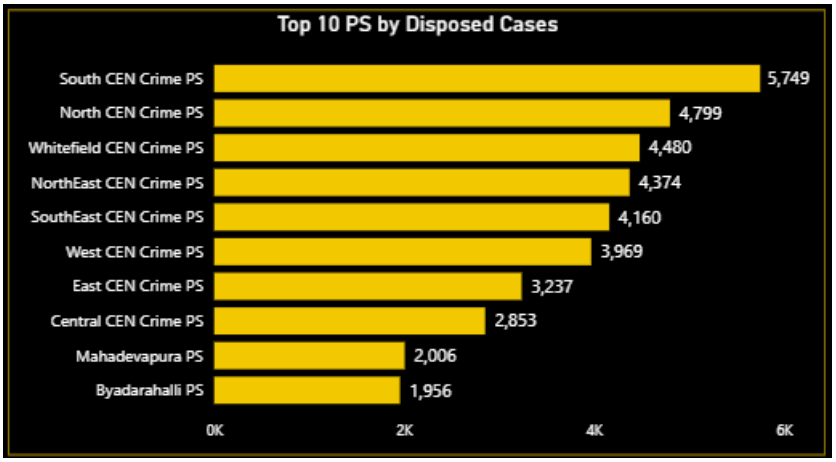
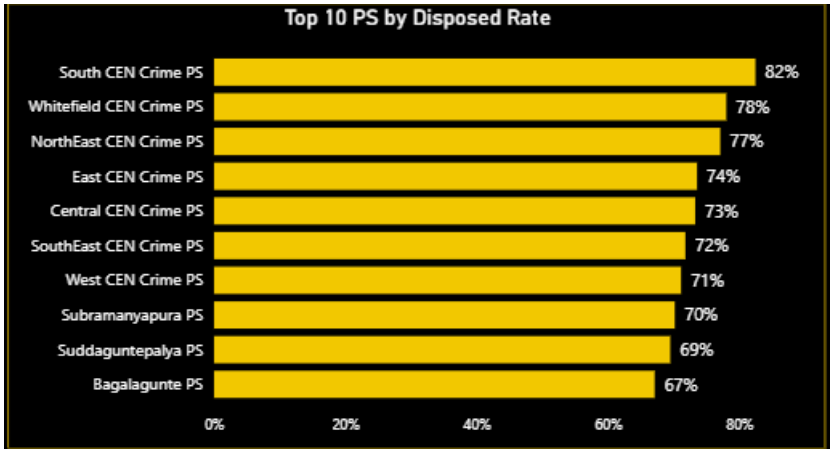


Top 10 PS by Disposed Cases

Aspect	Details
Visual Type	Clustered Bar Chart
Data Fields	**X-axis:** [Disposed Cases] **Y-axis:** UnitName **Filter:** UnitName Top10 by Disposal cases
Measures Used	[Disposed Cases]
Purpose	To track which Police Stations are disposing the most cases, showing productivity.
Insights	



Top 10 PS by Disposed Rate

Aspect	Details
Visual Type	Clustered Bar Chart
Data Fields	**X-axis:** [Disposed Rate] **Y-axis:** UnitName **Filter:** UnitName Top10 by Disposal Rate

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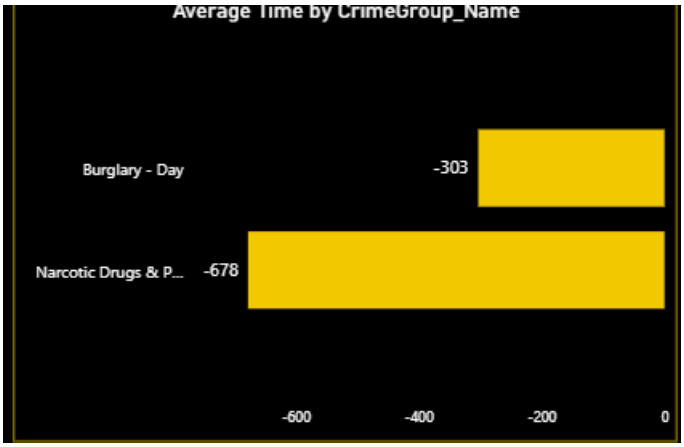
Measures Used	[Disposed Rate]
Purpose	To measure efficiency of Police Stations in clearing their assigned workload.
Insights	• South CEN PS has the highest disposal rate (82%). • Whitefield (78%), Northeast CEN (77%), and East CEN (74%) also show strong clearance efficiency. • Banashankari (44%) and Byadarahalli (56%) are weak performers.



Average Conviction Time by Crime Group

Aspect	Details
Visual Type	Clustered Bar Chart
Data Fields	**X-axis:** Average Conviction Time **Y-axis:** CrimeGroup_Name
Measures Used	[Average Conviction Time]
Purpose	To understand how long different crime groups take to reach conviction.
Insights	• NDPS (Narcotic Drugs) shows the longest average trial duration (-678 days, pending bias). • Burglary (Day) averages -303 days. • Indicates slow conviction in serious crime groups.

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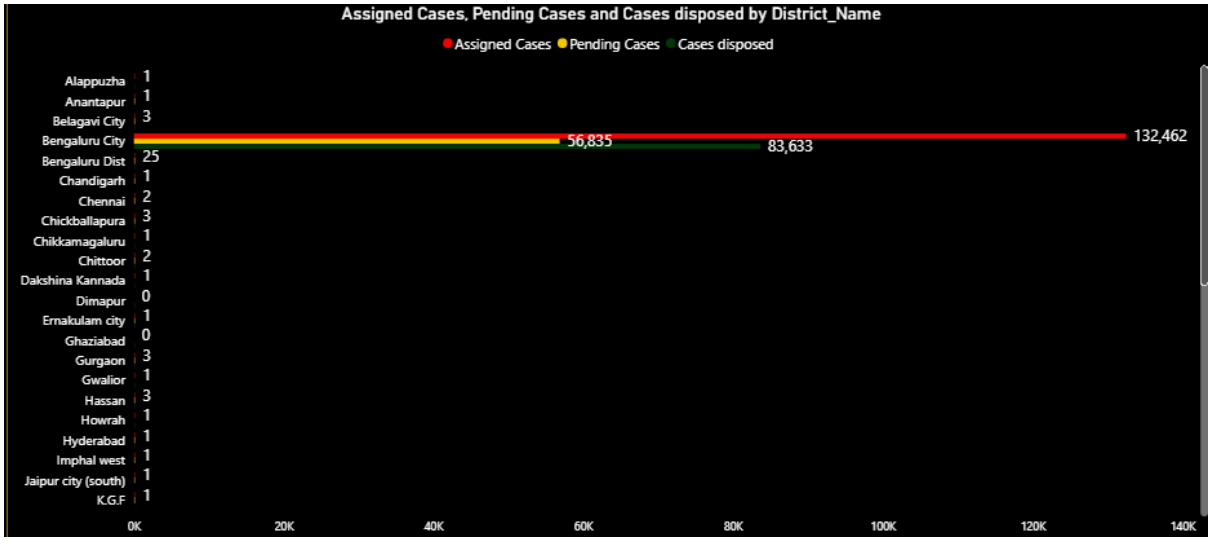
Problems / Limitations	• Negative values reflect calculation based on pending trials, so interpretation must be cautious. • Conviction time depends heavily on court process, not just PS work.
Recommendations	• Create dedicated NDPS fast-track courts. • Prioritize burglary and narcotics trials for quicker resolution. • Improve coordination between police & judiciary.



Assigned, Pending & Disposed Cases by District

Aspect	Details
Visual Type	Stacked Bar Chart
Data Fields	**X-axis**: [Assign Cases],[Pending Cases],[Disposed Cases] **Y-axis:** District Name
Measures Used	[Assigned Cases], [Pending Cases], [Disposed Cases]
Purpose	To compare workload and performance across districts.
Insights	• Bengaluru City dominates workload (132,462 assigned, 83,633 disposed, 56,835 pending). • Smaller districts (e.g., Alappuzha, Gwalior) have minimal case counts. • Imbalance is clear — urban hubs carry far higher burden.

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Problems / Limitations	• District size/population not factored — so absolute numbers may look skewed. - Does not show disposal efficiency by district.
Recommendations	• Normalize cases by population for fair comparison. • Provide extra manpower/resources to top-burden districts. • Regularly track district pendency vs. disposal ratio.

Challenges Faced During Dashboard Development

1. **Data Quality Issues**
   - **Observation:** Age and gender fields had missing values.
   - **Action:** Created calculated columns and bucketed age into segments (Child, Youth, Adult, Old).
2. **Partial Year Data (2025)**
   - **Observation:** Only half-year data available, causing misleading YoY results.
   - **Action:** Excluded 2025 from YoY logic and marked it clearly as YTD.
3. **Complex Navigation**
   - **Observation***:* Multiple buttons and bookmarks required for toggling visuals, making report management challenging.
   - **Action:** Organized bookmarks systematically and documented navigation.
4. **Performance Optimization**
   - **Observation***:* Bookmark-heavy visuals slowed report performance.
   - **Action:** Disabled unnecessary interactions and optimized visual layering.
5. **Spatial View Data Type Issue**
   - **Observation:** Crime hotspot map did not display all locations. Investigation showed longitude and latitude columns were stored as **Whole Number** instead of **Decimal**.
   - **Action***:* In Power Query, modified the transformation step and changed data type to Decimal. After correction, locations displayed correctly.
6. **Average Conviction Time Issue**
   - **Observation:** Average conviction time calculation returned blanks/zeros. Conviction Date column had missing values and incorrect data type/locale.
   - **Action:** In Power Query, changed data type to Date and adjusted locale (India). This resolved the issue, and valid conviction times were calculated.

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